

From Cross-Validation to SURE: Asymptotic Risk of Tuned Regularized Estimators

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Abstract

We derive the asymptotic risk function of regularized empirical risk minimization (ERM) estimators tuned by n -fold cross-validation (CV). The out-of-sample prediction loss of such estimators converges in distribution to the squared-error loss (risk function) of shrinkage estimators in the normal means model, tuned by Stein’s unbiased risk estimate (SURE). This risk function provides a more fine-grained picture of predictive performance than uniform bounds on worst-case regret, which are common in learning theory: it quantifies how risk varies with the true parameter.

As key intermediate steps, we show that (i) n -fold CV converges uniformly to SURE, and (ii) while SURE typically has multiple local minima, its global minimum is generically well separated. Well-separation ensures that uniform convergence of CV to SURE translates into convergence of the tuning parameter chosen by CV to that chosen by SURE.

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1 Introduction

Background The goal of supervised learning is to produce good predictions for new observations.¹ An important class of estimators for supervised learning can be described as regularized empirical risk minimization (ERM) estimators that are tuned using cross-validation (CV).

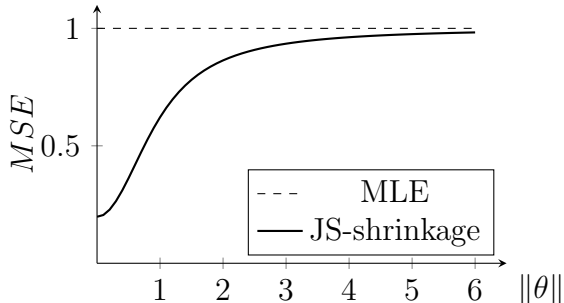
ERM estimators minimize in-sample average prediction loss (empirical risk), among a given class of predictors. Examples include ordinary least squares and maximum likelihood. ERM estimators are prone to overfitting if the class of predictors is large. Such estimators achieve low in-sample loss, but perform poorly for new observations. To counter overfitting, regularization is used. Regularization adds a penalty term to the ERM objective; common penalties include the L^2 norm of parameters (in Ridge regression) and the L^1 norm (in Lasso regression). Adding a penalty avoids overfitting, by reducing the estimator variance, at the cost of introducing some bias, which might result in underfitting.

To achieve good performance, avoiding both overfitting and underfitting, the amount of penalization needs to be carefully tuned. This can be done by choosing the weight on the penalty term as the minimizer of a cross-validation estimate of predictive loss. We focus on n -fold CV, where predictions are evaluated for one hold-out observation at a time, and predictive loss is estimated by averaging evaluations over each of the n observations.

Risk functions The present paper characterizes the behavior of estimators of this form by deriving an asymptotic approximation to their risk function. A large literature in learning theory characterizes such estimators by proving bounds on their worst-case regret—the supremum over data-generating processes (DGPs) of the difference between an estimator’s risk and the expected loss of the best predictor in the given class. Such bounds provide strong robustness guarantees, but they might not be informative about the behavior of predictive algorithms for realistic DGPs. By focusing on the risk function,

¹We would like to dedicate this paper to Gary Chamberlain, whose conversations provided the original inspiration for this project.

Figure 1: Risk function for JS-shrinkage, dimension 10



we obtain a more fine-grained characterization: the risk function tells us how expected predictive performance depends on the DGP, while worst-case regret bounds only characterize performance for the least favorable DGP.

Our asymptotic characterization relates the risk function of regularized ERM estimators tuned using CV to the risk function of the James-Stein (JS) shrinkage estimator James and Stein (1961), and generalizations thereof. JS shrinkage famously dominates maximum likelihood estimation (MLE) in the normal means setting: The risk function (mean squared error) of JS shrinkage, as characterized in Stein (1981), is lower than the risk of MLE, for any possible DGP. Figure 1 illustrates. Our result suggests that this same risk improvement carries over, asymptotically, to CV-tuned penalized estimation in general parametric models. Like CV, SURE provides an unbiased estimate of mean squared error; the JS estimator approximately minimizes SURE over the shrinkage intensity.

Main result Our main theorem states that the distribution of the out-of-sample prediction loss of CV-tuned regularized ERM estimators converges to the distribution of the squared-error loss of the corresponding SURE-tuned shrinkage estimator in the Gaussian limit experiment. In particular, the risk function (expected out-of-sample prediction loss, as a function of the true parameter) converges to the mean squared error of SURE-tuned penalized estimation in the normal means model. Two of our intermediate results are of

independent interest: the uniform approximation of n -fold CV by SURE, and the generic well-separation of the global minimum of SURE.

Key steps Let us briefly outline the three main parts of our proof. First, we show that in large samples ERM estimators are approximately normally distributed, and that out-of-sample predictive loss is approximately equal to squared error loss. These are standard results, and we follow van der Vaart (2000) in proving this step. We furthermore need to show that this approximation carries over to penalized estimators, for fixed tuning parameters. Our asymptotic approximations are based on local-to-0 asymptotics: As sample size n increases, the parameter vector drifts to 0 (which is the minimizer of the penalty term) at a rate of $1/\sqrt{n}$. This rate is such that both bias and variance remain non-negligible in the limit. The drifting-parameter framework is natural for studying regularized estimators, because it is the regime in which the penalty has a first-order effect: if the true parameter were fixed away from 0, the penalty would become asymptotically irrelevant, while if it were exactly 0, no bias-variance tradeoff would arise.

Second, we need to show that n -fold CV, as a random function that maps tuning parameter values to estimates of predictive loss, converges uniformly to SURE. In this step of the proof, we build on the prior work of Wilson et al. (2020). This step involves an influence function approximation for leave-one-out estimators, and a second-order approximation to predictive risk. Uniformity of convergence in the tuning parameter is key to this step, and we need to carefully specify regularity conditions such that uniformity is guaranteed.

Third, we need to show that convergence of the CV criterion function for tuning is sufficient for convergence of its minimizer, the tuned parameter. This is non-trivial, because both CV and SURE typically have multiple local minima, and might have multiple global minima. We need to show that generically the global minimum is well-separated. We do this using separate arguments for L^1 and L^2 penalties, characterizing the shape and behavior of SURE in either case.

We should emphasize some limitations of our analysis: We do not consider

penalties beyond L^1 and L^2 , or k -fold CV with $k < n$; extending our results in these directions is left for future work. Our asymptotic approximations are furthermore not appropriate in the over-parametrized regime, when the number of parameters is of similar or larger magnitude than the number of observations.

Literature The analysis in this paper connects several lines of work in statistics, econometrics, and machine learning. Leave-one-out cross-validation was formalized by Stone (1974); its asymptotic optimality for model selection was established by Li (1987). A closely related family of risk estimators includes Mallows’ C_p (Mallows, 1973), generalized cross-validation (Golub et al., 1979), and Stein’s unbiased risk estimate (Stein, 1981). Efron (2004) provides a unifying perspective, showing that these criteria all take the form of in-sample error plus a covariance penalty. Arlot and Celisse (2010) provide a comprehensive survey of cross-validation procedures and their theoretical properties. A noteworthy contrast with our results is provided by Shao (1993), who shows that leave-one-out CV is asymptotically *inconsistent* for model selection—it selects overfitted models with positive probability—while k -fold CV with $k = o(n)$ is consistent. Our result is complementary in focus: rather than asking which of a finite list of models has the best predictive ability, we characterize the limiting distribution and risk function of the estimator selected by leave-one-out CV, showing it converges to that of the SURE-tuned normal-means estimator.

Shrinkage estimation originates with James and Stein (1961) and was analyzed in depth by Stein (1981). SURE-based tuning of shrinkage was extended to wavelet thresholding by Donoho and Johnstone (1995). The two penalty families we study—Ridge (Hoerl and Kennard, 1970) and Lasso (Tibshirani, 1996)—are the most widely used forms of regularization in practice, and both are commonly tuned by cross-validation. The close relationship between leave-one-out CV and covariance-penalty criteria such as SURE is further illuminated by Zou et al. (2007), who show—using SURE as the analytical tool—that the effective degrees of freedom of the Lasso equals the number of nonzero fitted coefficients.

Using local asymptotic frameworks to characterize decision problems was pioneered by Le Cam (1972); work using this approach is reviewed in Hirano and Porter (2020). The use of shrinkage asymptotics for parametric models in econometrics is discussed in Hansen (2016). We build directly on Wilson et al. (2020), who provide non-asymptotic deterministic guarantees for approximate cross-validation, showing that leave-one-out estimators can be well approximated by a single Newton step from the full-sample estimator.

Roadmap The remainder of this paper is structured as follows: In Section 2, we introduce our model and assumptions, and define all relevant notation. In Section 3, we first provide a heuristic outline of our proof, and then state a series of intermediate lemmas. In Section 4, we illustrate our convergence results with several numerical experiments. Section 5 concludes. Proofs of all lemmas are collected in the appendix. In Appendix A, we verify the regularity conditions for leading examples. In Appendix B, we prove the lemmas corresponding to the first part of our argument, involving influence function approximations and asymptotic normality. In Appendix C, we prove the second part of the argument, namely the (uniform) approximation of n -fold CV by SURE. In Appendices D and E, we show that the global minimizer of SURE is generically well-separated for both L^2 and L^1 penalties, thereby proving the third part. In Appendix F, we conclude our derivation, proving the convergence of risk for tuned estimators.

2 Setup

In the following, we first set up our estimators, and their asymptotic counterparts, in a series of definitions. We then state the assumptions that will be invoked to justify our asymptotic approximations.

Throughout this paper, we consider the problem of estimating a parameter vector β_n which is defined as the minimizer of expected loss $E[l(\beta, Z_n^1)]$. For each n , the random vectors Z_n^i are i.i.d. draws from the distribution μ_n , across i . We use local-to-0 coordinates, where $\beta_n = \theta_0/\sqrt{n}$ for sample size

n , and we assume that θ_0 remains fixed across n . Any finite-sample object will be denoted by a subscript n ; objects without subscripts correspond to the limiting experiment.

For prediction problems, typically $Z_n^i = (W_n^i, Y_n^i)$, for predictive features W_n^i and outcomes Y_n^i . Examples include linear OLS regression, where $l(\beta, Z_n^i) = (Y_n^i - W_n^i \cdot \beta)^2$, as well as the use of neural nets² or other parametric models for classification, where $l(\beta, Z_n^i) = -\log(f(Y_n^i|W_n^i, \beta))$, and $f(Y_n^i|W_n^i, \beta)$ is the probability assigned to outcome Y_n^i by the model.

2.1 Definitions

Definition 1 introduces notation for loss functions and for limiting loss functions. We evaluate estimates of θ in terms of their expected loss $\bar{L}_n(\theta, \theta_0)$. For supervised learning, $\bar{L}_n(\theta, \theta_0)$ is the *out-of-sample* expected prediction error.

Definition 1 (Loss, empirical loss, and expected loss).

Given the loss function $l(\beta, z)$, define the following.

$$\begin{array}{ll}
l_n(\theta, z) = l(\theta/\sqrt{n}, z) & \text{Loss function in local parameter} \\
L_n(\theta) = \sum_{i=1}^n l_n(\theta, Z_n^i) & \text{Empirical loss} \\
\bar{L}_n(\theta, \theta_0) = E[L_n(\theta) - L_n(\theta_0)] & \text{Expected loss} \\
\bar{L}(\theta, \theta_0) = \lim_{n \rightarrow \infty} \bar{L}_n(\theta, \theta_0). & \text{Limiting expected loss}
\end{array}$$

We assume that the sequence of distributions μ_n is such that the limit in the last definition is well-defined.

Scaling Note that we could have equivalently defined

$$\bar{L}_n(\theta, \theta_0) = n \cdot E[l_n(\theta, Z_n^{n+1}) - l_n(\theta_0, Z_n^{n+1})],$$

²With a small number of parameters relative to the sample size.

that is, $\bar{L}_n(\theta, \theta_0)$ is the expected regret for out-of-sample predictions, multiplied by the sample size n . The multiplication by n is required because in Definition 1, we do *not* scale empirical loss $L_n(\theta)$ by a factor $\frac{1}{n}$, and therefore $L_n(\theta)$ diverges. In the definition of the local parameter vector θ we have however re-scaled the parameter vector β by a factor of $\frac{1}{\sqrt{n}}$. This implies that the Hessian (second derivative) of $L_n(\theta)$ with respect to θ , if it exists, is given by

$$\nabla_{\theta}^2 L_n(\theta) = \frac{1}{n} \sum_{i=1}^n \nabla_{\beta}^2 l(\theta/\sqrt{n}, Z_n^i).$$

We can thus expect that this Hessian converges, by a law of large numbers, under some additional regularity conditions.

Estimators of θ and their asymptotic counterparts We next specify a series of estimators, in Definition 2. We start with the standard ERM estimator $\hat{\theta}_n = \operatorname{argmin}_{\theta} L_n(\theta)$, and its limiting counterpart $\hat{\theta}$, where the latter is normally distributed with mean θ_0 and variance Σ . We then consider the regularized versions of these estimators, that is, the penalized ERM estimator with penalty $\lambda \cdot \pi(\theta)$, and its limiting counterpart. Our choice of local parametrization ensures that a constant value of λ along the sequence indexed by n leads to a non-degenerate limit for the penalized ERM estimator, where neither variance nor bias of this estimator vanish.

In Definition 2, we furthermore introduce leave-one-out loss, and the corresponding leave-one-out ERM and penalized ERM estimator. These will serve as the building blocks of n -fold cross-validation.

Definition 2 (Estimators of θ_0). *We will consider the following estimators of*

θ_0 , for finite n , and in the limit experiment:

$$\begin{aligned}\hat{\theta}_n &= \operatorname{argmin}_{\theta} L_n(\theta) && \text{ERM estimator} \\ \hat{\theta} &\sim N(\theta_0, \Sigma) && \text{Limiting ERM estimator} \\ \hat{\theta}_n^\lambda &= \operatorname{argmin}_{\theta} [L_n(\theta) + \lambda \cdot \pi(\theta)] && \text{Penalized ERM estimator} \\ \hat{\theta}^\lambda &= \operatorname{argmin}_{\theta} \left[\frac{1}{2} \|\theta - \hat{\theta}\|^2 + \lambda \cdot \pi(\theta) \right], && \text{Limiting penalized ERM estimator}\end{aligned}$$

where $\pi(\cdot)$ is convex and attains its minimum at 0.

We furthermore consider the following leave-one-out (LOO) loss and estimators of θ_0 :

$$\begin{aligned}L_n^{-i}(\theta) &= \sum_{j \neq i} l_n(\theta, Z_n^j) && \text{LOO empirical loss} \\ \hat{\theta}_n^{-i} &= \operatorname{argmin}_{\theta} L_n^{-i}(\theta) && \text{LOO ERM estimator} \\ \hat{\theta}_n^{\lambda, -i} &= \operatorname{argmin}_{\theta} [L_n^{-i}(\theta) + \lambda \cdot \pi(\theta)] . && \text{LOO penalized ERM estimator}\end{aligned}$$

We can rewrite the limiting penalized ERM estimator as

$$\hat{\theta}^\lambda = \hat{\theta} + g^\lambda(\hat{\theta}),$$

where

$$g^\lambda(\theta) = \operatorname{argmin}_g \frac{1}{2} \|g\|^2 + \lambda \cdot \pi(\theta + g).$$

Denote $\nabla g^\lambda(\theta)$ the derivative of $g^\lambda(\theta)$ where it exists, and define $\nabla g^\lambda(\theta) = 0$ at points where $g^\lambda(\theta)$ is not differentiable.³

Estimators of risk The preceding definition introduced penalized estimators for given, fixed values of the tuning parameter λ . We are interested in

³This convention is adopted for convenience; we will use it to handle Lasso (L^1) penalties in the proof of Lemma 4 below.

estimators which choose this tuning parameter in a data-dependent way, where λ minimizes an estimator of risk. For finite sample size n , we consider the n -fold crossvalidation (CV) criterion as an estimator of the risk of penalized ERM estimation. For the limit experiment, we consider Stein's Unbiased Risk Estimator (SURE) as an estimator of the risk of the penalized limiting ERM estimator.

Definition 3 (Estimators of risk). *We consider the following estimators of risk for penalized ERM estimators with fixed tuning parameter λ .*

$$CV_n(\lambda) = \sum_i l_n(\hat{\theta}_n^{\lambda, -i}, Z_n^i), \quad n\text{-fold CV}$$

$$SURE(\lambda, \hat{\theta}, \Sigma) = \frac{1}{2} \left[\text{trace}(\Sigma) + \|g^\lambda(\hat{\theta})\|^2 + 2 \text{trace} \left(\nabla g^\lambda(\hat{\theta}) \cdot \Sigma \right) \right]. \quad SURE$$

Tuned estimators We can now formally define our tuned estimators. $\hat{\theta}_n^*$ is the penalized ERM estimator using a tuning parameter λ_n^* which minimizes the n -fold CV estimator of risk. $\hat{\theta}^*$ is the penalized limiting ERM estimator using a tuning parameter λ^* which minimizes the SURE estimator of risk. The tuning parameter is chosen from a set $\Lambda \subset \mathbb{R}$. Later, we will consider $\Lambda = \mathbb{R}^+ \cup \{\infty\}$ (for Ridge penalties), and Λ arbitrary but finite (for Lasso penalties).

Definition 4 (Tuned estimators of θ).

$$\hat{\theta}_n^* = \hat{\theta}_n^{\lambda_n^*}, \quad \text{Penalized ERM tuned using CV}$$

$$\lambda_n^* = \underset{\lambda \in \Lambda}{\operatorname{argmin}} CV_n(\lambda)$$

$$\hat{\theta}^* = \hat{\theta}^{\lambda^*}, \quad \text{Limiting penalized ERM tuned using SURE}$$

$$\lambda^* = \underset{\lambda \in \Lambda}{\operatorname{argmin}} SURE(\lambda, \hat{\theta}, \Sigma).$$

We evaluate estimators based on their expected loss for new data-points. For supervised learning, this corresponds to the out-of-sample expected prediction loss. In this paper, we do not consider global criteria such as worst-case risk (maximizing over θ_0) or Bayes risk (averaging over a prior distribution

for θ_0). Instead, we are interested in the dependence of expected loss on the parameter θ_0 , which is captured by the risk function. Our notation makes this dependence on θ_0 explicit. The risk function gives a more fine-grained picture of estimator performance, relative to global criteria such as worst-case risk, Bayes risk, or worst-case regret.

The parameter θ_0 enters the following expressions both directly, as an argument of $\bar{L}_n$ and $\bar{L}$, and implicitly, via the distribution of Z that the expectations are averaging over.

Definition 5 (Risk functions).

$$\begin{aligned} R_n(\theta_0) &= E \left[\bar{L}_n(\hat{\theta}_n^*, \theta_0) \right] && \text{Finite sample risk, tuned using CV} \\ R(\theta_0) &= E \left[\bar{L}(\hat{\theta}^*, \theta_0) \right] && \text{Limiting risk, tuned using SURE.} \end{aligned}$$

2.2 Assumptions

Having defined our estimators and evaluation criteria, we next specify the assumptions invoked in our asymptotic analysis. Assumption 1 sets up a sequence of experiments, indexed by n . We assume that, for each n , the minimizer of expected loss $E[l(\beta, Z_n^i)]$ is given by $\theta_0/\sqrt{n}$. Put differently, the minimizer β_n of expected loss drifts towards 0. We furthermore assume that the variance Σ of the score $\nabla_\beta l(\theta_0/\sqrt{n}, Z_n^i)$ remains constant along our sequence.

Assumption 1 (Sequence of experiments). *For each n , the random vectors Z_n^i are i.i.d. draws from the distribution μ_n , across i . The distributions μ_n are such that θ_0 and Σ do not depend on n , where*

$$\begin{aligned} \theta_0 &= \operatorname{argmin}_{\theta} E[l(\theta/\sqrt{n}, Z_n^i)], \\ \Sigma &= \operatorname{Var} \left(\nabla_\beta l(\theta_0/\sqrt{n}, Z_n^i) \right). \end{aligned}$$

In particular, the minimizer $\beta_n = \theta_0/\sqrt{n}$ of expected loss drifts to 0 at rate $1/\sqrt{n}$. We furthermore assume that Σ is positive definite.

The limiting Hessian $H = \nabla_{\theta}^2 \bar{L}(\theta, \theta_0)$ is typically non-degenerate because of our scaling of $\bar{L}_n(\theta, \theta_0)$ and of θ . The following Assumption 2 is made for notational convenience. This assumption states that the Hessian H is equal to the identity I . This is a coordinate normalization that can be imposed without loss of generality.⁴

Assumption 2 (Normalized loss function).

$$\nabla_{\theta}^2 \bar{L}(\theta, \theta_0)|_{\theta=\theta_0} = I.$$

The last part of our proof requires showing that the global optimum of *SURE* with respect to λ is generically unique and well-separated. We will prove this fact for both Ridge and Lasso penalties, using separate arguments for either case.

Assumption 3 (Penalty function and grid for tuning).

The penalty $\pi(\theta)$ and the set Λ take one of the following two forms:

1. **Ridge:** $\pi(\theta) = \frac{1}{2} \theta \cdot A^{-1} \cdot \theta$, where A is positive definite, and $\Lambda = \mathbb{R}^+ \cup \{\infty\}$.
2. **Lasso:** $\pi(\theta) = \|A^{-1} \cdot \theta\|_1$, where A is an invertible matrix, and $\Lambda \subset \mathbb{R}^+$ is finite.

The remaining assumptions state regularity conditions. The first item in Assumption 4 is a condition on the loss function which allows us to invoke results from empirical process theory. Similar assumptions are invoked in van der Vaart (2000) when deriving the properties of M-estimators. The second item in Assumption 4 is a weak high-level condition ruling out divergence of ERM estimators, which ensures the applicability of empirical process results.

Assumption 4 (Conditions for convergence of the ERM estimator).

⁴By suitable choice of coordinates we can normalize *either* H , or the asymptotic variance Σ of Assumption 1, or the Hessian of the penalty function π (when the latter exists), but not more than one of these three matrices, in general. After normalizing the Hessian, we can however diagonalize one more matrix, without loss of generality.

1. **Lipschitz loss**

The loss function $l(\beta, z)$ satisfies

$$|l(\beta_1, z) - l(\beta_2, z)| \leq m(z) \cdot \|\beta_1 - \beta_2\|,$$

for all β_1, β_2 in a neighborhood of 0, where $\sup_n \text{Var}(m(Z_n^i)) \leq \infty$. Furthermore, $l(\beta, Z_n^i)$ is differentiable w.r.t. β , for all β in a neighborhood of $\beta = 0$, with probability 1.

2. **Stochastically bounded ERM estimator**

The sequence $\hat{\theta}_n = \text{argmin}_{\theta} L_n(\theta)$ is bounded in probability.

To prove the (uniform) convergence of CV_n to $SURE$, we finally impose the following additional regularity conditions.

Assumption 5 (Conditions for convergence of CV).

1. **Conditions on loss**

There exist $\mu > 0, \nu < \infty$ independent of n such that $L_n(\theta)$ is μ -strongly convex and has ν -smooth Hessians with probability approaching 1 under μ_n .⁵

2. **Conditions on scores**

The function $\sqrt{n}\nabla_{\theta}l_n(\theta, Z_n^i)$ is Lipschitz continuous almost everywhere, i.e., there exists $B_n(Z_n^i)$ such that

$$\|\sqrt{n}\nabla_{\theta}l_n(\theta, Z_n^i) - \sqrt{n}\nabla_{\theta}l_n(\theta', Z_n^i)\| \leq B_n(Z_n^i) \|\theta - \theta'\| \quad \forall \theta, \theta' \in \Theta$$

and $E_{\mu_n} [\|B_n(Z_n^i)\|^2] < \infty$. Additionally, there exists $M < \infty$ independent of n such that

$$\mathbb{E}_{\mu_n} [\|\sqrt{n}\nabla_{\theta}l_n(\theta_0, Z_n^i)\|^4] \leq M.$$

⁵ $L_n(\theta)$ is μ -strongly convex if $\nabla^2 L_n(\theta) - \mu I$ is positive semi-definite for all θ . A function is L -smooth if its gradients are Lipschitz continuous with Lipschitz constant L .

3. *Conditions on Hessians*

The Hessian $\nabla_{\theta}^2 l_n(\theta, Z_n^i)$ is such that

$$\frac{1}{n} \sum_{i=1}^n \left\| n \nabla_{\theta}^2 l_n(\theta_0, Z_n^i) \right\|^2 = O_{\mu_n}(1).$$

Furthermore, there exists $C_n(Z_n^i)$ such that

$$\left\| n \nabla_{\theta}^2 l_n(\theta, Z_n^i) - n \nabla_{\theta'}^2 l_n(\theta', Z_n^i) \right\| \leq C_n(Z_n^i) \|\theta - \theta'\| \quad \forall \theta, \theta' \in \Theta$$

and $\sup_n E_{\mu_n} [C_n(Z_n^i)^2] < \infty$.

4. *Conditions on Fourth Derivatives*

The fourth derivative tensor, $D_{\theta}^4 l_n(\theta, Z_n^i)$, of $l_n(\cdot, Z_n^i)$ is such that

$$\sup_n \mathbb{E}_{\mu_n} \left[\sup_{\theta \in \Theta} \left\| n^2 D_{\theta}^4 l_n(\theta, Z_n^i) \right\|^4 \right] \leq M,$$

for some $M < \infty$.

In Appendix A, we verify these assumptions for standard linear regression and generalized linear models.

3 Main result and intermediate lemmas

Our main goal in this paper is to prove the following result:

Theorem 1.

$$\bar{L}_n(\hat{\theta}_n^*, \theta_0) \rightarrow_d \frac{1}{2} \|\hat{\theta}^* - \theta_0\|^2.$$

In words, the distribution of loss of the penalized ERM estimator tuned using n-fold cross-validation converges to the distribution of squared error of the corresponding shrinkage estimator in the normal means model, tuned by Stein's Unbiased Risk Estimate. A special case of these limiting estimators are James-Stein shrinkage estimators, for which closed form characterizations

of the risk function are known (Stein, 1981). An immediate corollary of Theorem 1 is the convergence of risk functions, subject to possible truncation of tail events.⁶

Corollary 1. *Let $M > 0$. Then*

$$E \left[\min \left(\bar{L}_n(\hat{\theta}_n^*, \theta_0), M \right) \right] \rightarrow E \left[\min \left(\frac{1}{2} \|\hat{\theta}^* - \theta_0\|^2, M \right) \right].$$

3.1 Outline of proof

We will build up our argument that proves Theorem 1 in a series of lemmas. Before doing so, however, we first provide an intuitive sketch of our argument, while neglecting remainder terms. Subsequently, we will prove that these remainder terms are indeed asymptotically negligible.

Influence function approximation We start by noting that empirical risk is asymptotically equivalent to quadratic error loss relative to the sample mean $\tilde{\theta}_n$,

$$L_n(\theta) \approx \text{const.} + \frac{1}{2} \|\theta - \tilde{\theta}_n\|^2, \quad \tilde{\theta}_n = \theta_0 + \frac{1}{\sqrt{n}} \sum_i X_n^i,$$

where

$$X_n^i = -\nabla_{\beta} l(\theta_0 / \sqrt{n}, Z_n^i)$$

is the influence function. Recall that we have normalized the Hessian in Assumption 2, which simplifies the expression for $\tilde{\theta}_n$. This approximation of empirical risk immediately implies an asymptotic linear approximation of the empirical risk minimization (ERM) estimator, $\hat{\theta}_n \approx \tilde{\theta}_n$. These are standard approximations that deliver asymptotic normality of ERM estimators; see for instance Theorem 5.21 in van der Vaart (2000).

⁶Truncation is necessary because, even for estimators such as linear OLS regression, risk is typically undefined, since some eigenvalues of the design matrix might be close to 0, so that the moments of $\hat{\theta}_n$ might not exist.

We then get the corresponding approximation for the penalized ERM estimator, for fixed λ ,

$$\hat{\theta}_n^\lambda \approx \tilde{\theta}_n^\lambda = \tilde{\theta}_n + g^\lambda(\tilde{\theta}_n),$$

where we recall the definition $g^\lambda(\theta) = \operatorname{argmin}_g \frac{1}{2}\|g\|^2 + \lambda \cdot \pi(\theta + g)$.

Convergence of CV to SURE An analogous approximation holds for leave-one-out (LOO) loss. To obtain the LOO sample mean, the influence function $\frac{1}{\sqrt{n}}X_n^i$ is subtracted from the sample mean $\tilde{\theta}_n$, which gives

$$L_n^{-i}(\theta) \approx \text{const.} + \frac{1}{2}\|\theta - \tilde{\theta}_n^{-i}\|^2, \quad \text{where } \tilde{\theta}_n^{-i} = \tilde{\theta}_n - \frac{1}{\sqrt{n}}X_n^i.$$

The penalized LOO estimator is then approximately given by

$$\hat{\theta}_n^{\lambda,-i} \approx \tilde{\theta}_n^{-i} + g^\lambda(\tilde{\theta}_n^{-i}) \approx \tilde{\theta}_n^\lambda - \frac{1}{\sqrt{n}}(I + \nabla g^\lambda(\tilde{\theta}_n)) \cdot X_n^i.$$

In the last step we have replaced g^λ by its first-order Taylor approximation around $\tilde{\theta}_n$, at points $\tilde{\theta}_n$ where g^λ is differentiable. (This approximation won't hold at kink-points of g^λ , which exist for Lasso penalties, in particular.)

The n-fold cross-validation estimator of the risk of $\hat{\theta}_n^\lambda$ can be approximated by

$$CV_n(\lambda) = \sum_i l_n(\hat{\theta}_n^{\lambda,-i}, Z_n^i) \approx \text{const.} + \frac{1}{2n} \sum_i \|\hat{\theta}_n^{\lambda,-i} - \theta_0 - \sqrt{n}X_n^i\|^2.$$

When we take this expression, plug in the approximate form of $\hat{\theta}_n^{\lambda,-i}$, multiply

out the inner products, and omit terms which do not depend on λ , we obtain

$$\begin{aligned}
CV_n(\lambda) &\approx \text{const.} + \frac{1}{2n} \sum_i \underbrace{\left\| \tilde{\theta}_n + g^\lambda(\tilde{\theta}_n) - \frac{1}{\sqrt{n}}(I + \nabla g^\lambda(\tilde{\theta}_n)) \cdot X_n^i - \theta_0 - \sqrt{n}X_n^i \right\|^2}_{\approx \hat{\theta}_n^{\lambda, -i}} \\
&\approx \text{const.} + \frac{1}{2n} \sum_i \|g^\lambda(\tilde{\theta}_n)\|^2 + \frac{1}{n} \sum_i \langle \nabla g^\lambda(\tilde{\theta}_n) \cdot X_n^i, X_n^i \rangle \\
&\approx \text{const.} + \frac{1}{2} \|g^\lambda(\hat{\theta}_n)\|^2 + \text{trace}(\nabla g^\lambda(\hat{\theta}_n) \cdot \hat{\Sigma}_n) \\
&= \text{const.} + SURE(\lambda, \hat{\theta}_n, \hat{\Sigma}_n) \\
&\approx \text{const.} + SURE(\lambda, \hat{\theta}_n, \Sigma),
\end{aligned}$$

where $\hat{\Sigma}_n$ is the sample second moment of X_n . In the second line, *const.* subsumes any terms that do not depend on λ , while the approximation omits terms that depend on λ but are of order $1/\sqrt{n}$. This approximation to CV_n has the form of Stein's Unbiased Risk Estimate. The first term in this approximation to $CV_n(\lambda)$ corresponds to the average in-sample error, the second term has the form of a covariance penalty (Efron, 2004).

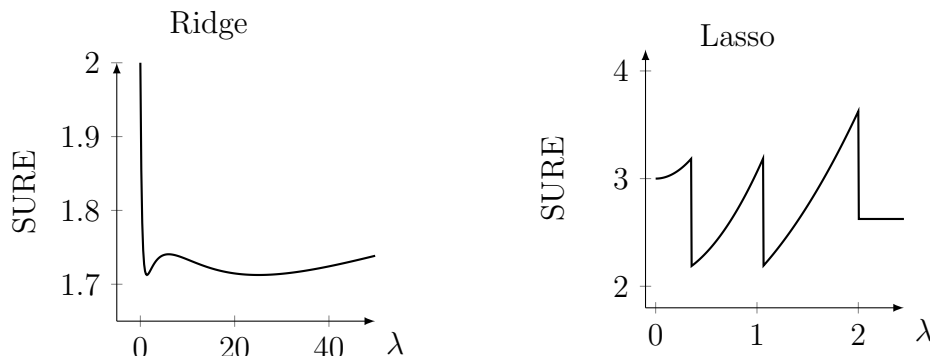
Convergence of tuning parameter and tuned estimators We will need to show that this approximation is uniformly valid in λ . We will furthermore need to show that uniform proximity of CV_n to $SURE$ is enough to guarantee proximity of the corresponding optimized tuning parameters,

$$\argmin_{\lambda} CV_n(\lambda) \approx \argmin_{\lambda} SURE(\lambda, \hat{\theta}_n, \hat{\Sigma}_n).$$

This latter step is non-trivial, because the criterion function $SURE(\lambda, \hat{\theta}_n, \hat{\Sigma}_n)$ typically has multiple local minima. For certain values of $\hat{\theta}_n$ this function furthermore has multiple *global* minima. When $SURE$ has multiple global (near-)minima, uniform closeness of CV to $SURE$ is not enough to ensure closeness of the minimizer of CV to the minimizer of $SURE$. The plots in Figure 2, which are reproduced from Wilson et al. (2020),⁷ illustrate two nu-

⁷The numerical values corresponding to these examples are as follows: (a) $SURE$ for Ridge: $\hat{\theta} = (1.3893, 1.5)$, $L(\theta) = (\theta - \hat{\theta}) \text{diag}(1, 40)(\theta - \hat{\theta})$, $\pi(\theta) = \|\theta\|^2$, (b) $SURE$ for Lasso:

Figure 2: Examples of multi-modality of *SURE*



Notes: These plots show examples of multi-modality for *SURE*, for the case of L^2 penalties (Ridge) and L^1 penalties (Lasso). Both examples are reproduced from Wilson et al. (2020).

merical examples (realizations of $\hat{\theta}_n$ and of $\hat{\Sigma}_n$) for which *SURE* indeed has multiple global minima.

Using separate arguments for Ridge (Appendix D) and Lasso (Appendix E), we will prove, however, that the global minimum of *SURE* with respect to λ is unique and well separated almost everywhere, in a suitable sense. Put differently, cases such as those represented in Figure 2 are non-generic, such that they do not lead to a breakdown of convergence for the optimized tuning parameter. The arguments proving that multiple global minima only occur on a set of measure 0 are the most non-standard part of our proof

Well-separation ensures that the argmin functional is continuous at almost every realization of $\hat{\theta}$. From these results we thus conclude that the mapping from $\hat{\theta}_n$ and CV_n to the tuned estimate $\hat{\theta}_n^*$ is almost everywhere continuous. This allows us to invoke the continuous mapping and dominated convergence theorems, and to conclude the proof of Theorem 1.

3.2 Intermediate lemmas

Let us now turn to a more formal exposition of our argument. We will prove Theorem 1 in a series of Lemmas. The lemmas are stated in this section,

$$\hat{\theta} = \frac{1}{\sqrt{n}}(\sqrt{1/8}, \sqrt{9/8}, 2), \pi(\theta) = \sum |\theta_j|.$$

Table 1: Assumptions invoked by each result.

Result	Assumptions invoked	Builds on
Lemma 1 (Lipschitz g^λ)	convexity of π	—
Lemma 2 (Influence-function approx.)	1, 2, 4.1, 4.2, 5.2	Lemma 1
Lemma 3 (Limiting squared-error loss)	1, 2, 5.3	—
Corollary 2 (Asymptotic distribution)	1, 4.1	Lemmas 1, 2
Lemma 4 (CV $\rightarrow$ SURE)	1, 2, 3, 5.1–5.4	Lemmas 1, 2, Cor. 2
Lemma 5 (Tuning-parameter convergence)	1, 3	Lemmas 2, 4, Cor. 2
Theorem 1 (Risk convergence)	all	Lemmas 3, 5
Corollary 1 (Convergence of expected loss)	all	Theorem 1

their proofs in the appendices. All results impose the assumptions stated in Section 2. For transparency, Table 1 records which assumptions each result invokes directly, beyond those inherited through the results it builds on.

Lemma 1 (Lipschitz g^λ). *For any $\lambda \geq 0$, if $\pi(\cdot)$ is convex then $g^\lambda(\theta) = \operatorname{argmin}_g \frac{1}{2}\|g\|^2 + \lambda \cdot \pi(\theta + g)$ is Lipschitz with Lipschitz constant 1.*

Lemma 2 (Influence function approximation).

$$L_n(\theta) - L_n(\theta_0) = \frac{1}{2}\|\theta - \tilde{\theta}_n\|^2 - \frac{1}{2}\|\tilde{\theta}_n - \theta_0\|^2 + \epsilon_n(\theta), \quad (1)$$

where $\sup_{\theta: \|\theta\| \leq C} \epsilon_n(\theta) = o_{\mu_n}(1)$ and $\sup_{\theta: \|\theta\| \leq C} \nabla \epsilon_n(\theta) = o_{\mu_n}(1)$ for any $C < \infty$, and

$$\tilde{\theta}_n = \theta_0 + \frac{1}{\sqrt{n}} \sum_i X_n^i, \quad X_n^i = -\nabla_{\beta} l(\theta_0/\sqrt{n}, Z_n^i).$$

The ERM and penalized ERM estimators

$$\hat{\theta}_n = \operatorname{argmin}_{\theta} L_n(\theta), \quad \hat{\theta}_n^\lambda = \operatorname{argmin}_{\theta} [L_n(\theta) + \lambda \cdot \pi(\theta)]$$

satisfy

$$\hat{\theta}_n = \tilde{\theta}_n + o_{\mu_n}(1), \quad \sup_{\lambda} \|\hat{\theta}_n^\lambda - \tilde{\theta}_n - g^\lambda(\tilde{\theta}_n)\| = o_{\mu_n}(1).$$

Lemma 3 (Limiting squared error loss). *The limiting expected loss is well defined and given by*

$$\bar{L}(\theta, \theta_0) = \frac{1}{2} \|\theta - \theta_0\|^2.$$

Convergence of $\bar{L}_n(\theta, \theta_0)$ to this limit is uniform in any bounded neighborhood of θ_0 : $\sup_{\theta: \|\theta - \theta_0\| \leq C} |\bar{L}_n(\theta, \theta_0) - \bar{L}(\theta, \theta_0)| \rightarrow 0$ for all $C < \infty$.

Corollary 2 (Asymptotic distribution for fixed tuning parameter). *The ERM and penalized ERM estimators satisfy*

$$\hat{\theta}_n \rightarrow^d \hat{\theta} \sim N(\theta_0, \Sigma), \quad \hat{\theta}_n^\lambda \rightarrow^d \hat{\theta} + g^\lambda(\hat{\theta}).$$

Lemma 4 (Convergence of CV to SURE). *There exists a λ -independent (data-dependent) constant c_n such that the n -fold crossvalidation criterion satisfies*

$$\sup_{\lambda \in \Lambda} \left| CV_n(\lambda) - c_n - SURE(\lambda, \hat{\theta}_n, \Sigma) \right| \xrightarrow{\mu_n} 0.$$

Lemma 5 (Joint convergence of tuning parameter and tuned estimators).

$$(\lambda_n^*, \hat{\theta}_n) \rightarrow^d (\lambda^*, \hat{\theta}).$$

and

$$\hat{\theta}_n^* \rightarrow^d \hat{\theta}^*.$$

From Lemma 5, we then show Theorem 1. The appendices prove each of these Lemmas in turn.

Discussion of assumptions Let us briefly discuss the role of our different assumptions, as referenced in Table 1, in ensuring the validity of these results. Each assumption controls a distinct source of error in our approximations. Assumption 1 places the problem in a local-to-zero regime, where the truth drifts at rate $1/\sqrt{n}$ while the score variance Σ stays fixed and non-degenerate; this is what keeps the bias-variance tradeoff of penalization non-trivial in the limit, so that CV and $SURE$ estimate a meaningful, non-degenerate risk.

Assumption 2 normalizes the limiting Hessian to the identity; it carries no statistical content and merely puts the limit experiment into the normal-means form on which *SURE* is built. Assumption 3 restricts attention to Ridge and Lasso penalties, the two cases for which we can show that the *SURE* objective has a unique, well-separated minimizer in λ almost everywhere—the property that lets convergence of the *criteria* carry over to convergence of the *tuned* estimators (Lemma 5). The two conditions in Assumption 4 govern the unregularized ERM estimator: the Lipschitz condition (4.1) lets us invoke empirical-process arguments for its convergence, while the stochastic-boundedness condition (4.2) rules out runaway estimates that would invalidate those arguments. Finally, the four conditions in Assumption 5 provide the quantitative control needed to equate CV_n with *SURE*: strong convexity and smooth Hessians (5.1) make each leave-one-out refit a small, $O(n^{-1/2})$ perturbation, so the degrees-of-freedom correction is well-behaved, and the score, Hessian, and fourth-derivative moment bounds (5.2–5.4) control the successive terms in the Taylor expansion of the leave-one-out loss, ensuring the remainder vanishes uniformly in λ .

4 Numerical experiments

We now illustrate our results numerically. The experiments have two purposes: to verify that the approximations underlying our intermediate lemmas are accurate at realistic sample sizes, and to show that the risk of a cross-validated estimator indeed converges to the risk of its *SURE*-tuned limit. Replication code is available at https://github.com/maxkasy/cv_and_sure_simulations.

Designs We compare each finite-sample experiment to its limiting normal-means counterpart. In the finite-sample experiment we draw n i.i.d. observations, form the empirical risk minimizer $\hat{\theta}_n$, the n -fold cross-validation criterion $CV_n(\lambda)$, the CV-tuned estimator $\hat{\theta}_n^*$, and its out-of-sample loss $\bar{L}_n(\hat{\theta}_n^*, \theta_0)$. In the limiting experiment we draw $\hat{\theta} \sim N(\theta_0, \Sigma)$ and compute the *SURE*-tuned estimator $\hat{\theta}^*$ and its loss $\frac{1}{2}\|\hat{\theta}^* - \theta_0\|^2$.

All designs are constructed directly in the normalized coordinates of Assumption 2, so that the limiting Hessian equals the identity, $H = I$. We consider the same examples discussed in Appendix A - linear regression and logisitic regression, for both Ridge and Lasso penalties:

For *linear regression*, $l(\beta, z) = (y - w \cdot \beta)^2$ with $W \sim N(0, \frac{1}{2}I_p)$ and $Y = W \cdot \theta_0/\sqrt{n} + U$, $U \sim N(0, \frac{1}{2})$; this yields $H = 2 E[WW^\top] = I$ and $\Sigma = I$, and the expected out-of-sample loss is exactly $\bar{L}_n(\theta, \theta_0) = \frac{1}{2}\|\theta - \theta_0\|^2$. For *logistic regression*, $l(\beta, z) = -\log f(y \mid w, \beta)$ with $W \sim N(0, 4I_p)$ and $Y \sim \text{Bernoulli}(\Lambda(W \cdot \theta_0/\sqrt{n}))$, Λ the logistic c.d.f.; here $b''(0) = \frac{1}{4}$, so again $H = I$ and $\Sigma = I$. The Ridge penalty is $\pi(\theta) = \frac{1}{2}\theta \cdot A^{-1}\theta$ and the Lasso penalty $\pi(\theta) = \|A^{-1}\theta\|_1$, with A diagonal. Table 2 lists the five resulting designs that we consider.

All five designs use $p = 10$. Since $\Sigma = I$, each coordinate of $\hat{\theta}$ has unit noise scale, and we choose the signal θ_0 to be of comparable, order-one magnitude. This is the regime in which the bias-variance tradeoff is non-degenerate: Some shrinkage strictly improves on both the unregularized estimator ($\lambda = 0$) and full shrinkage ($\lambda = \infty$), so that the tuning problem is not trivially solved at an endpoint. For the dense designs (D1, D3) all coordinates carry the same unit signal; for the Lasso designs (D2, D4) the signal is sparse, with three strong nonzero entries against a background of exact zeros, which is the regime the Lasso is designed for. Design D5 is the anisotropic stress test: its penalty matrix A has a wide, geometric eigenvalue spread ($A_{jj} = 40^{(j-1)/9}$, ranging from 1 to 40), the higher-dimensional analogue of the anisotropic Ridge configuration that produced the multi-modal *SURE* examples of Figure 2 in $p = 2$. Genuine multi-modality of *SURE* is a low-dimensional, knife-edge phenomenon that does not survive in $p = 10$; what the anisotropy produces here is a *SURE* objective that is poorly separated (nearly flat) near its minimum for a sizeable fraction of draws. This probes the well-separation argument behind Lemma 5 precisely where it is most demanding.

For each design we vary the sample size n and average over Monte Carlo replications. The n -fold cross-validation criterion is computed exactly: in closed form via leave-one-out residuals for the linear-Ridge designs (D1, D5),

Table 2: Simulation designs.

	Loss	Penalty	p	θ_0	A
D1	Linear	Ridge	10	$(1, \dots, 1)$	I
D2	Linear	Lasso	10	$(3, -3, 2, 0, \dots, 0)$	I
D3	Logit	Ridge	10	$(1, \dots, 1)$	I
D4	Logit	Lasso	10	$(3, -3, 2, 0, \dots, 0)$	I
D5	Linear	Ridge	10	$1.5 \cdot (1, \dots, 1)$	$\text{diag}(40^{(j-1)/9})$

$\Sigma = I$ and $p = 10$ throughout (normalized coordinates). The sparse signal is $\theta_0 = (3, -3, 2, 0, \dots, 0)$; for D5, $A = \text{diag}(40^0, 40^{1/9}, \dots, 40^1)$ is a geometric grid spanning $[1, 40]$.

and by explicit leave-one-out refitting otherwise. We use a grid of 14 values of λ . For Ridge, the grid is equally spaced in the compactified coordinate $s = \lambda/(1 + \lambda) \in [0, 1]$, ranging from $\lambda = 0$ (no shrinkage, $s = 0$) to $\lambda = \infty$ (full shrinkage, $s = 1$), consistent with the compactified $\Lambda = \mathbb{R}^+ \cup \{\infty\}$ of Assumption 3. For Lasso, the grid is equally spaced in $[0, \lambda_{\max}]$ with $\lambda_{\max} = 3$.

Diagnostics We summarize each approximation by a single scale-free statistic, reported in the columns of Table 3, and write each as a discrepancy Δ with a descriptive subscript. Let $\tilde{\theta}_n$ denote the influence-function approximation of Lemma 2.

1. **Influence-function approximation** (Lemma 2): $\Delta_{\text{IF}} = E\|\hat{\theta}_n - \tilde{\theta}_n\|^2 / E\|\hat{\theta}_n - \theta_0\|^2$, the mean squared error of the linearization relative to the scale of the estimand.
2. **Convergence of CV to SURE** (Lemma 4):

$$\Delta_{\text{CV}} = \frac{E \sup_{\lambda} |[CV_n(\lambda) - CV_n(\lambda_{\text{ref}})] - [SURE(\lambda) - SURE(\lambda_{\text{ref}})]|}{E [\max_{\lambda} SURE - \min_{\lambda} SURE]},$$

the uniform gap between the two criteria, recentered at a reference λ_{ref} to remove the λ -independent constant c_n , and normalized by the range of $SURE$ over the grid. Here $SURE(\lambda)$ is evaluated at the realized $\hat{\theta}_n$ and the true Σ .

Table 3: Approximation diagnostics at $n = 400$.

Design		Δ_{IF} (Lem. 2)	Δ_{CV} (Lem. 4)	Δ_{tune} (Lem. 5)	R_n (Thm. 1)	R (limit)	Δ_R $ R_n - R $
D1	Linear / Ridge	0.028	0.208	0.052	3.110	3.084	0.026
D2	Linear / Lasso (sparse)	0.028	0.226	0.111	3.885	3.789	0.096
D3	Logit / Ridge	0.032	0.309	0.055	3.062	3.075	0.013
D4	Logit / Lasso (sparse)	0.038	0.297	0.127	3.731	3.776	0.046
D5	Linear / Ridge (anisotropic)	0.028	0.152	0.036	4.827	4.733	0.094

Columns: Δ_{IF} (Lemma 2), Δ_{CV} (Lemma 4), Δ_{tune} (Lemma 5), the finite-sample and limiting risks R_n, R , and their gap Δ_R (Theorem 1). Averages over 8000 replications for the closed-form designs D1, D5 and 2000 for the leave-one-out designs D2–D4.

- Agreement of the tuning parameter** (Lemma 5): $\Delta_{\text{tune}} = E|\ell(\lambda_n^*) - \ell(\lambda^S)| / E\ell(\lambda^S)$, where $\ell(\lambda) = \frac{1}{2}\|\hat{\theta}_n + g^\lambda(\hat{\theta}_n) - \theta_0\|^2$ and λ_n^*, λ^S minimize CV_n and $SURE(\cdot, \hat{\theta}_n, \Sigma)$ at the same $\hat{\theta}_n$. This is the relative excess loss incurred by tuning with CV rather than $SURE$ — directly addressing whether the two minimizers are close enough to matter.
- Convergence of risk** (Theorem 1, Corollary 1): $\Delta_R = |R_n - R|$, the gap between the finite-sample risk $R_n = E[\min(\bar{L}_n(\hat{\theta}_n^*, \theta_0), M)]$ and its limit $R = E[\min(\frac{1}{2}\|\hat{\theta}^* - \theta_0\|^2, M)]$, with truncation $M = 20$.

Results Table 3 reports the four diagnostics at $n = 400$, and Figures 3 and 4 display their behavior as n grows. All four diagnostics are small across every design, and Figure 3 shows that they vanish as n grows. The influence-function error Δ_{IF} decreases at the parametric rate n^{-1} — for example from 0.22 at $n = 50$ to 0.014 at $n = 800$ for design D1, halving with each doubling of n — confirming the linearization $\hat{\theta}_n \approx \tilde{\theta}_n$ of Lemma 2. The cross-validation gap Δ_{CV} decreases at the slower rate $n^{-1/2}$, as expected for a quantity governed by the sampling variability of the leave-one-out perturbations; the convergence holds uniformly across the Ridge and Lasso penalties and across the linear and logistic losses, confirming Lemma 4.

The tuning diagnostic Δ_{tune} shows that selecting λ by cross-validation rather than by $SURE$ costs only a few percent of the tuned loss, and shrinks

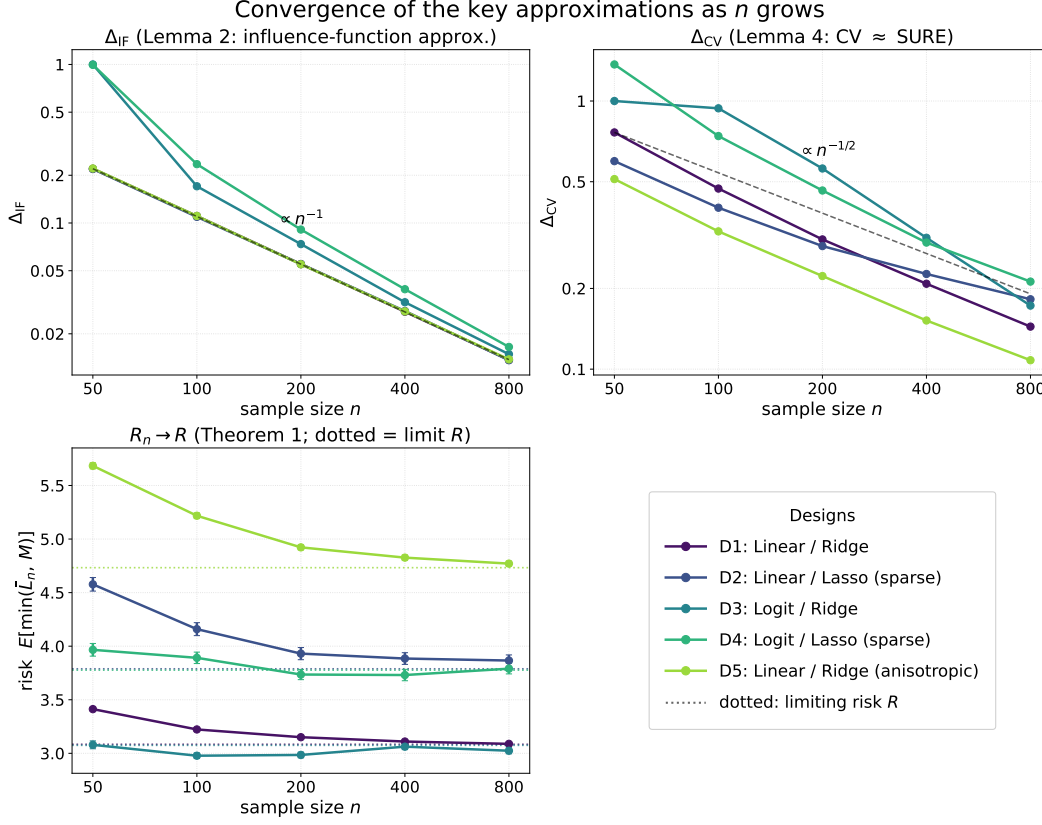


Figure 3: Convergence of the approximations as n grows (designs identified in the legend panel, lower right). Top left: the influence-function error Δ_{IF} (Lemma 2), $\propto n^{-1}$. Top right: the cross-validation gap Δ_{CV} (Lemma 4), $\propto n^{-1/2}$. Bottom left: the finite-sample risk R_n converging to the limit R (dotted lines; Theorem 1).

toward zero as n grows. It is largest for the two Lasso designs (D2, D4: around 0.11–0.13 at $n = 400$), whose non-quadratic, kinked objective is the least smooth, and smallest for the anisotropic design D5 (0.036). The latter is instructive: the wide eigenvalue spread of D5 makes *SURE* nearly flat near its minimum, so although the minimizer is poorly separated, the loss is correspondingly insensitive to the exact choice of λ . Poor separation and large excess loss thus need not coincide. Consequently the finite-sample risk R_n is close to its limit R already at $n = 400$ (the gap Δ_R is at most a fraction of a unit, against risk levels of 3 to 5), and converges to it as n grows; by $n = 800$ the two are nearly indistinguishable.

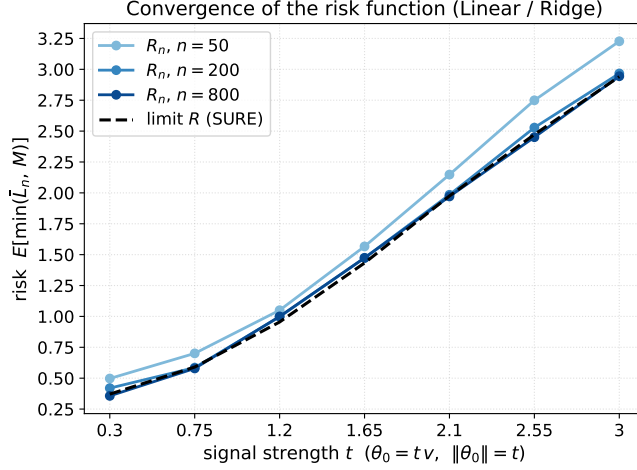


Figure 4: Convergence of the risk function (linear–Ridge design). Along the slice $\theta_0 = t \cdot v$ (with v a fixed unit vector, so $\|\theta_0\| = t$), the finite-sample risk $R_n(\theta_0)$ of the CV-tuned estimator approaches the SURE-tuned limit $R(\theta_0)$; by $n = 200$ the curves are nearly indistinguishable from the limit.

Figure 4 makes the convergence concrete for the linear–Ridge design along a one-dimensional slice $\theta_0 = t \cdot v$: the finite-sample risk function $R_n(\theta_0)$ approaches the limiting risk function $R(\theta_0)$ predicted by Theorem 1, and the two are nearly indistinguishable by $n = 200$. This is the practical content of our result: the equivalence between cross-validation and SURE is useful at realistic sample sizes, not only asymptotically.

Note also that the limiting risk $R(\theta_0)$ rises from about 0.4 near the origin toward—but always below—the risk of the *unregularized* estimator: at $\lambda = 0$ the limiting estimator is $\hat{\theta} \sim N(\theta_0, \Sigma)$, whose risk is θ_0 -independent and equal to $E[\frac{1}{2}\|\hat{\theta} - \theta_0\|^2] = \frac{1}{2} \text{trace}(\Sigma) = p/2 = 5$. The CV- and SURE-tuned estimators thus strictly improve on the unregularized estimator across the entire slice, and indeed in every design of Table 3, where the tuned limiting risks range from 3.1 to 4.7—all below 5. This confirms the well-known dominance of James–Stein shrinkage over the maximum-likelihood estimator (Figure 1).

5 Conclusion

We have shown that the out-of-sample prediction loss of regularized empirical risk minimization estimators tuned by n -fold cross-validation converges in distribution to the squared-error loss of the corresponding shrinkage estimator in the normal-means model, tuned by Stein’s unbiased risk estimate. We conclude by discussing what this equivalence tells us.

For *practitioners*, the result justifies using SURE as an asymptotically equivalent and computationally cheaper surrogate for n -fold cross-validation: SURE is available in closed form for both the Ridge and the Lasso penalty, and it avoids refitting the model n times. Beyond this computational gain, the limiting risk function provides a uniform, robust performance guarantee for estimators tuned using CV.

For *theorists*, our result reduces the analysis of CV-tuned regularized ERM to a shrinkage problem in the normal-means model. This yields a characterization of the asymptotic *risk function*, which describes how risk varies with the underlying parameter θ_0 . This provides a rich characterization: Instead of a single uniform upper bound on regret, as in standard learning theory, the risk function allows to distinguish the parameter regions where regularization helps more or less. It also implies that the well-developed apparatus for the normal-means model (Stein’s identity, James-Stein shrinkage, etc.) can be used to analyze the asymptotic behavior of other data-driven tuning procedures.

Our analysis has some limitations that suggest directions for future work. Our framework is in particular fixed-dimensional and local-to-zero: the number of parameters is held fixed as n grows, and the optimal parameter drifts to zero at the $\sqrt{n}$ rate. The high-dimensional case, in which the dimension grows with the sample size, is likely to behave differently. This high-dimensional case might provide a more accurate description of machine learning practice. In many such regimes one would expect the data-driven tuning parameter to be *consistent* for its optimal value (as, for example, in Abadie and Kasy 2019); the variability of the tuning parameter could then be ignored, and the limiting risk function would simplify accordingly.

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A Verifying the regularity conditions for leading examples

We now verify that our assumptions hold for two leading examples: linear regression with quadratic error loss, and smooth generalized linear models. Throughout, $Z_n^i = (W_n^i, Y_n^i)$ with regressors $W_n^i \in \mathbb{R}^p$ i.i.d across n and i , and we write $Q = E[W_n^i W_n^{i\top}]$.

Assumptions 1–3 are easy to verify for the following examples: Assumption 1 holds immediately if an appropriate local-to-zero data-generating process is specified (with $\Sigma \succ 0$ and constant in n checked explicitly), Assumption 2 is the coordinate normalization of the Hessian (e.g. $2Q = I$ for linear regression), and Assumption 3 is simply the choice of Ridge or Lasso penalty. We therefore focus on verifying the regularity conditions in Assumptions 4 and 5.

Linear regression. Let $l(\beta, z) = (y - w \cdot \beta)^2$, with data generated as $Y_n^i = W_n^i \cdot \beta_n + U_n^i$, where $\beta_n = \theta_0/\sqrt{n}$, $E[U_n^i \mid W_n^i] = 0$, and $E[(U_n^i)^2 \mid W_n^i] \geq \sigma^2 > 0$. Assume $Q \succ 0$ and the uniform moment bounds $E\|W_n^i\|^8 \leq M$ and $E[(U_n^i)^8] \leq M$. The score and Hessian are

$$\nabla_\beta l(\beta, z) = -2w(y - w \cdot \beta), \quad \nabla_\beta^2 l(\beta, z) = 2ww^\top,$$

so the influence function is $X_n^i = -\nabla_\beta l(\beta_n, Z_n^i) = 2W_n^i U_n^i$, with $\Sigma = \text{Var}(X_n^i) = 4E[W_n^i W_n^{i\top} (U_n^i)^2] \succ 0$, as required by Assumption 1. The Hessian is constant in β with expectation $2Q$, and Assumption 2 is the coordinate normalization $2Q = I$, imposed without loss of generality.

Assumption 4. For item 1, on any ball $\|\beta\| \leq \delta$ we have $\|\nabla_\beta l(\beta, z)\| \leq 2\|w\|(|y| + \|w\|\delta) =: m(z)$, and $\sup_n \text{Var}(m(Z_n^i)) < \infty$ follows from the moment bounds (using $|Y_n^i| \leq \|W_n^i\| \|\beta_n\| + |U_n^i|$ with $\|\beta_n\| \rightarrow 0$); the loss is everywhere differentiable. For item 2, the ERM estimator is OLS, $\hat{\theta}_n = \theta_0 + (\frac{1}{n} \sum_i W_n^i W_n^{i\top})^{-1} \frac{1}{\sqrt{n}} \sum_i W_n^i U_n^i = \theta_0 + O_{\mu_n}(1)$, since $\frac{1}{n} \sum_i W_n^i W_n^{i\top} \rightarrow Q \succ 0$ and $\frac{1}{\sqrt{n}} \sum_i W_n^i U_n^i = O_{\mu_n}(1)$ by the central limit theorem; hence $\hat{\theta}_n$ is bounded

in probability.

Assumption 5. Recall $l_n(\theta, z) = l(\theta/\sqrt{n}, z)$, so $\sqrt{n}\nabla_\theta l_n = \nabla_\beta l$, $n\nabla_\theta^2 l_n = \nabla_\beta^2 l$, and $n^2 D_\theta^4 l_n = D_\beta^4 l$, all evaluated at $\beta = \theta/\sqrt{n}$.

- *Item 1.* $\nabla_\theta^2 L_n(\theta) = \frac{1}{n} \sum_i 2W_n^i W_n^{i\top} \rightarrow 2Q \succ 0$, with eigenvalues bounded above and below with probability approaching one; hence L_n is μ -strongly convex and ν -smooth for suitable $0 < \mu < \nu$.
- *Item 2.* As $\nabla_\beta l$ is affine in β , $\|\sqrt{n}\nabla_\theta l_n(\theta, z) - \sqrt{n}\nabla_\theta l_n(\theta', z)\| = \|2ww^\top(\theta - \theta')/\sqrt{n}\| \leq (2\|w\|^2/\sqrt{n})\|\theta - \theta'\|$, so $B_n(z) = 2\|w\|^2/\sqrt{n}$ with $E[B_n(Z_n^i)^2] = 4E\|W_n^i\|^4/n < \infty$. Moreover $E\|\sqrt{n}\nabla_\theta l_n(\theta_0, Z_n^i)\|^4 = 16E[\|W_n^i\|^4(U_n^i)^4] \leq 16\sqrt{E\|W_n^i\|^8 E[(U_n^i)^8]} \leq 16M$.
- *Item 3.* $n\nabla_\theta^2 l_n(\theta, z) = 2ww^\top$ is constant in θ , so $\frac{1}{n} \sum_i \|2W_n^i W_n^{i\top}\|^2 = 4 \cdot \frac{1}{n} \sum_i \|W_n^i\|^4 = O_{\mu_n}(1)$, and the Lipschitz bound holds with $C_n(z) \equiv 0$.
- *Item 4.* The loss is quadratic, so $D_\beta^4 l \equiv 0$ and the bound holds with $M = 0$.

Smooth generalized linear models. Let $l(\beta, z) = -\log f(y | w, \beta)$ for an exponential-family model with $\log f(y | w, \beta) = y(w \cdot \beta) - b(w \cdot \beta) + c(y)$ and log-partition function $b(\cdot)$, as in logistic or Poisson regression. Then $\nabla_\beta l = -(y - b'(w \cdot \beta))w$, $\nabla_\beta^2 l = b''(w \cdot \beta) ww^\top$, and $D_\beta^k l = (-1)^k b^{(k)}(w \cdot \beta) w^{\otimes k}$. Suppose the regressors are bounded, $\|W_n^i\| \leq B$ almost surely, and $\sup_n E[(Y_n^i)^4] \leq M$. The log partition function satisfies $b'' > 0$, so the loss is strictly convex; near the drifting truth $\beta_n \rightarrow 0$ the curvature $b''(W \cdot \beta_n)$ is bounded below, giving $\nabla_\theta^2 L_n(\theta) \rightarrow E[b''(0)W_n^i W_n^{i\top}] \succ 0$ and hence Assumption 5(i). Because b is smooth with derivatives bounded on the compact set $\{|w \cdot \beta| \leq B\delta\}$ and the regressors are bounded, the score, Hessian, and fourth-derivative bounds (items 2–4) reduce to moments of $\|W_n^i\|$, which are finite by boundedness; Assumption 4 follows as above. (For logistic regression $b(t) = \log(1 + e^t)$, with $0 < b'' \leq 1/4$ and all $b^{(k)}$ bounded.)

B Proofs: Influence function approximations

B.1 Proof of Lemma 1 (Lipschitz g^λ)

Fix $0 \leq \lambda < \infty$. Recall that $g^\lambda(\theta) = \operatorname{argmin}_g \frac{1}{2}\|g\|^2 + \lambda \cdot \pi(\theta + g)$. If $\pi(\cdot)$ is convex, then so is the objective function on the right. This implies that there exists a sub-gradient $\nabla\pi$ of π such that the first order condition

$$g + \lambda \cdot \nabla\pi(\theta + g) = 0$$

holds for $g = g^\lambda(\theta)$. Consider two values θ_1, θ_2 of θ , and the corresponding solutions g_1, g_2 and sub-gradients $\nabla\pi_1, \nabla\pi_2$, as well as the differences $\Delta\theta = \theta_2 - \theta_1$ and $\Delta g = g_2 - g_1$. Taking the difference of the first order condition across the two values yields.

$$\Delta g + \lambda \cdot [\nabla\pi_2 - \nabla\pi_1] = 0.$$

Convexity of π implies

$$\langle \nabla\pi_2 - \nabla\pi_1, \Delta\theta + \Delta g \rangle \geq 0.$$

Combining the last two equations yields

$$\langle \Delta g, \Delta\theta + \Delta g \rangle = -\lambda \cdot \langle \nabla\pi_2 - \nabla\pi_1, \Delta\theta + \Delta g \rangle \leq 0,$$

and thus (using Cauchy-Schwartz to get the second inequality),

$$\|\Delta g\|^2 \leq \langle \Delta g, -\Delta\theta \rangle \leq \|\Delta g\| \cdot \|\Delta\theta\|,$$

so that

$$\|\Delta g\| \leq \|\Delta\theta\|.$$

This proves that $g^\lambda(\theta)$ is Lipschitz with Lipschitz-constant 1. □

B.2 Proof of Lemma 2 (Influence function approximation)

Assume first that Equation (1) holds, so that $L_n(\theta) - L_n(\theta_0) = \frac{1}{2}\|\theta - \tilde{\theta}_n\|^2 - \frac{1}{2}\|\tilde{\theta}_n - \theta_0\|^2 + \epsilon_n(\theta)$. We show, under this assumption, that $\sup_\lambda \|\hat{\theta}_n^\lambda - \tilde{\theta}_n^\lambda\|^2 = o_{\mu_n}(1)$. Leveraging convexity of π and Lipschitz continuity of g^λ , we first bound the corresponding difference in penalized squared error, which allows us to bound the difference in squared error, and finally the difference between the estimators themselves.

Bounding the difference in penalized squared error loss Define

$$\tilde{\theta}_n^\lambda = \operatorname{argmin}_\theta \left[\frac{1}{2}\|\theta - \tilde{\theta}_n\|^2 + \lambda \cdot \pi(\theta) \right] = \tilde{\theta}_n + g^\lambda(\tilde{\theta}_n).$$

By definition,

$$\hat{\theta}_n^\lambda = \operatorname{argmin}_\theta [L_n(\theta) + \lambda \cdot \pi(\theta)]$$

and thus

$$L_n(\hat{\theta}_n^\lambda) + \lambda \cdot \pi(\hat{\theta}_n^\lambda) \leq L_n(\tilde{\theta}_n^\lambda) + \lambda \cdot \pi(\tilde{\theta}_n^\lambda).$$

Substituting for $L_n(\cdot)$ on both sides of this inequality, using Equation (1) applied to both $\theta = \hat{\theta}_n^\lambda$ and $\theta = \tilde{\theta}_n^\lambda$, and rearranging yields

$$\left[\frac{1}{2}\|\hat{\theta}_n^\lambda - \tilde{\theta}_n\|^2 + \lambda \cdot \pi(\hat{\theta}_n^\lambda) \right] - \left[\frac{1}{2}\|\tilde{\theta}_n^\lambda - \tilde{\theta}_n\|^2 + \lambda \cdot \pi(\tilde{\theta}_n^\lambda) \right] \leq \epsilon_n(\tilde{\theta}_n^\lambda) - \epsilon_n(\hat{\theta}_n^\lambda). \quad (2)$$

Bounding the difference in squared error loss We next prove the following claim: Convexity of π , and the definition $\tilde{\theta}_n^\lambda = \operatorname{argmin}_\theta \left[\frac{1}{2}\|\theta - \tilde{\theta}_n\|^2 + \lambda \cdot \pi(\theta) \right]$, imply that, for any θ ,

$$\frac{1}{2}\|\theta - \tilde{\theta}_n^\lambda\|^2 \leq \left[\frac{1}{2}\|\theta - \tilde{\theta}_n\|^2 + \lambda \cdot \pi(\theta) \right] - \left[\frac{1}{2}\|\tilde{\theta}_n^\lambda - \tilde{\theta}_n\|^2 + \lambda \cdot \pi(\tilde{\theta}_n^\lambda) \right]. \quad (3)$$

To show (3), denote $a(\theta) = \frac{1}{2}\|\theta - \tilde{\theta}_n\|^2$ and $b(\theta) = \lambda \cdot \pi(\theta)$. We can write

$$\frac{1}{2}\|\theta - \tilde{\theta}_n^\lambda\|^2 = a(\theta) - a(\tilde{\theta}_n^\lambda) - \nabla a(\tilde{\theta}_n^\lambda) \cdot (\theta - \tilde{\theta}_n^\lambda).$$

By convexity of π and optimality of $\tilde{\theta}_n^\lambda$, there exists a subgradient ∇b of b such that

$$\nabla a(\tilde{\theta}_n^\lambda) + \nabla b(\tilde{\theta}_n^\lambda) = 0.$$

Eliminating the common terms $a(\theta) - a(\tilde{\theta}_n^\lambda)$ on the left and right hand side, we can now rewrite (3) as

$$-\nabla a(\tilde{\theta}_n^\lambda) \cdot (\theta - \tilde{\theta}_n^\lambda) \leq b(\theta) - b(\tilde{\theta}_n^\lambda).$$

But since $-\nabla a(\tilde{\theta}_n^\lambda) = \nabla b(\tilde{\theta}_n^\lambda)$, this inequality holds by convexity of b and the definition of a subgradient, and the claim follows.

Bounding the distance between estimators Combining two inequalities (2) and (3) yields

$$\frac{1}{2} \|\hat{\theta}_n^\lambda - \tilde{\theta}_n^\lambda\|^2 \leq \epsilon_n(\tilde{\theta}_n^\lambda) - \epsilon_n(\hat{\theta}_n^\lambda).$$

It follows from Assumption 4 item 2, which states that $\hat{\theta}_n^\lambda$ is bounded in probability, and the Lipschitzness of g^λ (Lemma 1), which implies $\|\tilde{\theta}_n^\lambda\| \leq 2\|\hat{\theta}_n^\lambda\|$, that both $\hat{\theta}_n^\lambda$ and $\tilde{\theta}_n^\lambda$ are bounded in probability. Combined with equation (1), we consequently obtain that with probability approaching 1 under μ_n , there exists $C < \infty$ such that

$$\sup_\lambda \frac{1}{2} \|\hat{\theta}_n^\lambda - \tilde{\theta}_n^\lambda\|^2 \leq \left(2 \sup_{\|\theta\| \leq C} \epsilon_n(\theta) \right) = o(1).$$

The statement for the ERM estimator follows as a special case, where $\lambda = 0$.

Proving Equation (1), using empirical process theory It remains to show that (1) holds, where $\sup_{\|\theta\| \leq C} \epsilon_n(\theta) = o_{\mu_n}(1)$. This claim follows from a straightforward generalization of the proof of Lemma 19.31 in van der Vaart (2000) to the case of drifting distributions.

Applicability of arguments of Lemma 19.31 in van der Vaart (2000) is guaranteed by the conditions in Assumption 4, item 1. In particular, pointwise

convergence, for fixed θ , follows from almost sure differentiability of l , by dominated convergence, given the uniform bound on the variance of $m(Z_n^i)$ in Assumption 4. To get uniform convergence across values of θ in any ball of radius δ around 0, tightness needs to be shown. Tightness follows from a bound on the bracketing number of the class of functions $\{\sqrt{n}(l_n(\theta, \cdot) - l_n(0, \cdot)) : \|\theta\| \leq \delta\}$. The bound in the proof of Lemma 19.31 in van der Vaart (2000) applies verbatim, with a constant C that does not depend on n , based on the uniform bound on the variance of $m(Z_n^i)$ in Assumption 4. The claim follows.

The claim that $\sup_{\theta: \|\theta\| \leq C} \nabla \epsilon_n(\theta) = o_{\mu_n}(1)$ follows from the same argument, applied to $\nabla_\theta l_n(\theta, Z_n^i)$, using the condition on scores in item 2 of Assumption 5. $\square$

B.3 Proof of Lemma 3 (Limiting squared error loss)

By Definition 1, $\bar{L}_n(\theta, \theta_0) = E[L_n(\theta) - L_n(\theta_0)]$, and $\bar{L}_n(\theta, \theta_0)$ is minimized at $\theta = \theta_0$. By a second order Taylor expansion around $\theta = \theta_0$,

$$\bar{L}_n(\theta, \theta_0) = \frac{1}{2}(\theta - \theta_0) \cdot \nabla_\theta^2 \bar{L}_n(\tilde{\theta}, \theta_0) \cdot (\theta - \theta_0).$$

for some $\tilde{\theta}$ between θ and θ_0 . By Assumption 2, $\nabla_\theta^2 \bar{L}(\theta, \theta_0)|_{\theta=\theta_0} = I$. By definition,

$$\nabla_\theta^2 \bar{L}_n(\theta, \theta_0) = \nabla_\beta^2 E[l(\beta, Z_n^i)]|_{\beta=\theta/\sqrt{n}}.$$

The claim of the lemma then follows from continuity of the Hessian of $\nabla_\beta^2 E[l(\beta, Z_n^i)]$ at $\beta = 0$. Continuity of the Hessian follows from item 3 of Assumption 5:

$$\begin{aligned} & \left\| \nabla_\beta^2 E[l(\beta, Z_n^i)]|_{\beta=\theta/\sqrt{n}} - \nabla_\beta^2 E[l(\beta, Z_n^i)]|_{\beta=0} \right\| \\ & \leq E \left[\left\| \nabla_\beta^2 l(\beta, Z_n^i)|_{\beta=\theta/\sqrt{n}} - \nabla_\beta^2 l(\beta, Z_n^i)|_{\beta=0} \right\| \right] \\ & \leq E[C_n(Z_n^i)] \cdot \frac{\|\theta\|}{\sqrt{n}}, \end{aligned}$$

where $\sup_n E[C_n(Z_n^i)] < \infty$; this follows from $\sup_n E[C_n(Z_n^i)^2] < \infty$ (Assumption 5.3) via Jensen's inequality.

□

B.4 Proof of Corollary 2 (Asymptotic distribution for fixed tuning parameter)

Recall that $\text{Var}(X_n^1) = \Sigma$ is constant in n , by assumption. Note furthermore that Assumption 4 (item 1) implies the Lindeberg condition

$$E [\|X_n^1\|^2 \cdot \mathbf{1}(\|X_n^1\| > \sqrt{n}M)] \rightarrow 0$$

for all $M > 0$, since $\|X_n^1\| \leq m(Z_n^1)$: $X_n^i = -\nabla_{\beta} l(\theta_0/\sqrt{n}, Z_n^i)$, and the Lipschitz condition in Assumption 4.1 together with a.e. differentiability implies $\|\nabla_{\beta} l(\beta, Z_n^i)\| \leq m(Z_n^i)$ at all points of differentiability, and the variance of the latter is uniformly bounded. The Lindeberg-Feller central limit theorem (Proposition 2.27 in van der Vaart 2000), applied to the triangular array (X_n^i) , therefore implies $\tilde{\theta}_n \rightarrow^d N(\theta_0, \Sigma)$.

The claims of Corollary 2 then follow from Lemma 2, and the continuous mapping theorem, where continuity of g^{λ} follows from convexity of π , by Lemma 1.

□

C Proof of Lemma 4

Step 0 (Preliminary observations):

We start by stating some useful results for the proof. First, note that by Lemma 1 in Wilson et al. (2020) and Assumption 5(i), it follows

$$\sup_{\lambda} \left\| \hat{\theta}_n^{\lambda, -i} - \hat{\theta}_n^{\lambda} \right\| = O_{\mu_n} \left(\frac{1}{\mu} \left\| \nabla_{\theta} l_n(\hat{\theta}_n, Z_n^i) \right\| \right) = O_{\mu_n}(n^{-1/2}). \quad (4)$$

An analogous argument implies

$$\sup_{\lambda} \left\| \tilde{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^{\lambda} \right\| = O_{\mu_n}(n^{-1/2}), \quad (5)$$

where

$$\tilde{\theta}^{\lambda, -i} := \underset{\theta}{\operatorname{argmin}} \left[\frac{1}{2} \left\| \theta - \tilde{\theta}_n^{-i} \right\|^2 + \lambda \cdot \pi(\theta) \right]. \quad (6)$$

Second, recall from Lemma 2 that

$$\sup_{\lambda} \left\| \hat{\theta}_n^{\lambda} - \tilde{\theta}_n^{\lambda} \right\| = o_{\mu_n}(1). \quad (7)$$

The next set of results concern the properties of $g^{\lambda}(\cdot)$. Since $g^{\lambda}(\cdot)$ is Lipschitz continuous by Lemma 1, it is differentiable almost everywhere (by Rademacher's theorem). In particular, there exists an $R^{\lambda}(\cdot; \theta)$ such that

$$g^{\lambda}(\theta + \delta) = g^{\lambda}(\theta) + \nabla g^{\lambda}(\theta)^{\top} \delta + R^{\lambda}(\delta; \theta), \quad (8)$$

and

$$\lim_{\|\delta\| \rightarrow 0} \frac{\|R^{\lambda}(\delta; \theta)\|}{\|\delta\|} = 0 \text{ for each } \lambda \text{ and (Lebesgue) almost every } \theta. \quad (9)$$

In fact, under Assumption 3, we can strengthen (9) to:

$$\lim_{\|\delta\| \rightarrow 0} \sup_{\lambda \in \Lambda} \frac{\|R^\lambda(\delta; \theta)\|}{\|\delta\|} = 0 \text{ for (Lebesgue) almost every } \theta. \quad (10)$$

For Ridge, (10) is immediate, since $R^\lambda(\delta, \theta) = 0$. For Lasso, it follows from Lemma 8, which implies $R^\lambda(\delta, \theta) = 0$ for δ small enough, except on a set of θ values with Lebesgue measure 0.

For values of θ where $\nabla g^\lambda(\theta)$ does not exist, we somewhat arbitrarily set $\nabla g^\lambda(\theta) = 0$ and define $R^\lambda(\delta; \theta) = g^\lambda(\theta + \delta) - g^\lambda(\theta)$ for these values; that way (8) always holds. Observe that due to Lemma 1, $\|g^\lambda(\theta + \delta) - g^\lambda(\theta)\| \leq \|\delta\|$ and $\|\nabla g^\lambda(\theta)\| \leq 1$ (whenever the gradient exists), so

$$\sup_{\lambda, \theta} \|R^\lambda(\delta; \theta)\| \leq 2 \|\delta\|. \quad (11)$$

Step 1:

We first show that

$$\text{CV}_n(\lambda) = \sum_{i=1}^n l_n(\hat{\theta}_n^\lambda, Z_n^i) + \frac{1}{\sqrt{n}} \sum_{i=1}^n \left\langle \tilde{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^\lambda, \sqrt{n} \nabla_{\theta} l_n(\tilde{\theta}_n^\lambda, Z_n^i) \right\rangle + o_{\mu_n}(1), \quad (12)$$

uniformly over λ . By a first order Taylor expansion,

$$\begin{aligned} \text{CV}_n(\lambda) &= \sum_{i=1}^n l_n(\hat{\theta}_n^{\lambda, -i}, Z_n^i) \\ &= \sum_{i=1}^n l_n(\hat{\theta}_n^\lambda, Z_n^i) + \frac{1}{\sqrt{n}} \sum_{i=1}^n \left\langle \hat{\theta}_n^{\lambda, -i} - \hat{\theta}_n^\lambda, \sqrt{n} \nabla_{\theta} l_n(\hat{\theta}_n^\lambda, Z_n^i) \right\rangle \\ &\quad + \frac{1}{\sqrt{n}} \sum_{i=1}^n r_n(\hat{\theta}_n^{\lambda, -i}, \hat{\theta}_n^\lambda, Z_n^i), \end{aligned}$$

where

$$\left| r_n(\hat{\theta}_n^{\lambda, -i}, \hat{\theta}_n^\lambda, Z_n^i) \right| \leq B_n(Z_n^i) \cdot \left\| \hat{\theta}_n^{\lambda, -i} - \hat{\theta}_n^\lambda \right\|^2$$

by Assumption 5. Hence, by the Cauchy-Schwarz inequality,

$$\begin{aligned} & \left| \frac{1}{\sqrt{n}} \sum_{i=1}^n r_n \left(\hat{\theta}_n^{\lambda, -i}, \hat{\theta}_n^{\lambda}, Z_n^i \right) \right| \\ & \leq \left(\frac{1}{n} \sum_{i=1}^n |B_n(Z_n^i)|^2 \right)^{1/2} \left(\sum_{i=1}^n \left\| \hat{\theta}_n^{\lambda, -i} - \hat{\theta}_n^{\lambda} \right\|^4 \right)^{1/2}. \end{aligned}$$

The first term on the right hand side of the above expression is $O_{\mu_n}(1)$ by Assumption 5, while the second term is $O_{\mu_n}(n^{-1/2})$ by (4) and the two requirements of Assumption 5 since

$$\begin{aligned} & \sup_{\lambda} \sum_{i=1}^n \left\| \hat{\theta}_n^{\lambda, -i} - \hat{\theta}_n^{\lambda} \right\|^4 \\ & \leq \frac{1}{n^2 \mu^4} \sum_{i=1}^n \left\| \sqrt{n} \nabla_{\theta} l_n(\hat{\theta}_n, Z_n^i) \right\|^4 \\ & \leq \frac{8}{n^2 \mu^4} \sum_{i=1}^n \left\| \sqrt{n} \nabla_{\theta} l_n(\hat{\theta}_n, Z_n^i) - \sqrt{n} \nabla_{\theta} l_n(\theta_0, Z_n^i) \right\|^4 + \frac{8}{n^2 \mu^4} \sum_{i=1}^n \left\| \sqrt{n} \nabla_{\theta} l_n(\theta_0, Z_n^i) \right\|^4 \\ & = O_{\mu_n}(n^{-1}), \end{aligned} \tag{13}$$

so the expression overall is $O_{\mu_n}(n^{-1/2})$.

We now show that, uniformly over λ , one can approximate

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \left\langle \hat{\theta}_n^{\lambda, -i} - \hat{\theta}_n^{\lambda}, \sqrt{n} \nabla_{\theta} l_n \left(\hat{\theta}_n^{\lambda}, Z_n^i \right) \right\rangle. \tag{14}$$

with

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \left\langle \tilde{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^{\lambda}, \sqrt{n} \nabla_{\theta} l_n \left(\tilde{\theta}_n^{\lambda}, Z_n^i \right) \right\rangle.$$

To this end, we first argue that (14) can be approximated with

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \left\langle \hat{\theta}_n^{\lambda, -i} - \hat{\theta}_n^{\lambda}, \sqrt{n} \nabla_{\theta} l_n \left(\tilde{\theta}_n^{\lambda}, Z_n^i \right) \right\rangle.$$

By the Cauchy-Schwarz inequality, the approximation error is bounded by

$$\sup_{\lambda \in \Lambda} \left(\sum_{i=1}^n \left\| \hat{\theta}_n^{\lambda, -i} - \hat{\theta}_n^\lambda \right\|^2 \right)^{1/2} \cdot \sup_{\lambda \in \Lambda} \left(\frac{1}{n} \sum_{i=1}^n \left\| \sqrt{n} \nabla_{\theta} l_n \left(\hat{\theta}_n^\lambda, Z_n^i \right) - \sqrt{n} \nabla_{\theta} l_n \left(\tilde{\theta}_n^\lambda, Z_n^i \right) \right\|^2 \right)^{1/2}. \quad (15)$$

The first term in (15) is $O_{\mu_n}(1)$ by (4) and Assumption 5 (the argument is analogous to 13). The second term in (15) is $o_{\mu_n}(1)$ under (7) and Assumption 5(ii).

It then remains to show

$$\sup_{\lambda \in \Lambda} \left| \frac{1}{\sqrt{n}} \sum_{i=1}^n \left\langle \hat{\theta}_n^{\lambda, -i} - \hat{\theta}_n^\lambda, \sqrt{n} \nabla_{\theta} l_n \left(\tilde{\theta}_n^\lambda, Z_n^i \right) \right\rangle - \frac{1}{\sqrt{n}} \sum_{i=1}^n \left\langle \tilde{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^\lambda, \sqrt{n} \nabla_{\theta} l_n \left(\tilde{\theta}_n^\lambda, Z_n^i \right) \right\rangle \right| = o_{\mu_n}(1).$$

By the Cauchy-Schwarz inequality, the expression on the left is bounded by

$$\sup_{\lambda \in \Lambda} \left(\sum_{i=1}^n \left\| \left(\hat{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^{\lambda, -i} \right) - \left(\hat{\theta}_n^\lambda - \tilde{\theta}_n^\lambda \right) \right\|^2 \right)^{1/2} \cdot \sup_{\lambda} \left(\frac{1}{n} \sum_{i=1}^n \left\| \sqrt{n} \nabla_{\theta} l_n \left(\tilde{\theta}_n^\lambda, Z_n^i \right) \right\|^2 \right)^{1/2}.$$

The second term in the above expression is $O_{\mu_n}(1)$ by Assumption 5(ii) (4th moment bound on scores) and the Lipschitz condition, since $\tilde{\theta}_n^\lambda$ is bounded in probability. At the end of this proof, we analyze the first term, showing that

$$\sup_{\lambda \in \Lambda} \left(\sum_{i=1}^n \left\| \left(\hat{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^{\lambda, -i} \right) - \left(\hat{\theta}_n^\lambda - \tilde{\theta}_n^\lambda \right) \right\|^2 \right) = o_{\mu_n}(1).$$

Combining the above results proves (12).

Step 2:

Next, we show that uniformly over λ , the term

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \left\langle \tilde{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^\lambda, \sqrt{n} \nabla_{\theta} l_n \left(\tilde{\theta}_n^\lambda, Z_n^i \right) \right\rangle \quad (16)$$

in (12) can be approximated by

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \left\langle \tilde{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^\lambda, -X_n^i \right\rangle.$$

Indeed, under Assumption 5(iii), we have

$$\begin{aligned} & \frac{1}{\sqrt{n}} \sum_{i=1}^n \left\langle \tilde{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^\lambda, \sqrt{n} \nabla_{\theta} l_n \left(\tilde{\theta}_n^\lambda, Z_n^i \right) \right\rangle - \frac{1}{\sqrt{n}} \sum_{i=1}^n \left\langle \tilde{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^\lambda, -X_n^i \right\rangle \\ &= \frac{1}{n} \sum_{i=1}^n \left\langle \tilde{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^\lambda, n \nabla_{\theta}^2 l_n \left(\theta_0, Z_n^i \right) \cdot \left(\tilde{\theta}_n^\lambda - \theta_0 \right) \right\rangle + \frac{1}{n} \sum_{i=1}^n \left\langle \tilde{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^\lambda, \Delta_n \left(\tilde{\theta}_n^\lambda - \theta_0, Z_n^i \right) \right\rangle, \end{aligned} \quad (17)$$

where

$$\left\| \Delta_n \left(\tilde{\theta}_n^\lambda - \theta_0, Z_n^i \right) \right\| \leq C_n(Z_n^i) \cdot \left\| \tilde{\theta}_n^\lambda - \theta_0 \right\|^2,$$

and the function $C_n(\cdot)$ is defined in Assumption 5(iii).

By the Cauchy-Schwarz inequality, Assumption 5 and (5),

$$\begin{aligned} & \sup_{\lambda \in \Lambda} \left| \frac{1}{n} \sum_{i=1}^n \left\langle \tilde{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^\lambda, n \nabla_{\theta}^2 l_n \left(\theta_0, Z_n^i \right) \cdot \left(\tilde{\theta}_n^\lambda - \theta_0 \right) \right\rangle \right| \\ & \leq n^{-1/2} \cdot \sup_{\lambda \in \Lambda} \left(\sum_{i=1}^n \left\| \tilde{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^\lambda \right\|^2 \right)^{1/2} \cdot \sup_{\lambda \in \Lambda} \left(\frac{1}{n} \sum_{i=1}^n \left\| n \nabla_{\theta}^2 l_n \left(\theta_0, Z_n^i \right) \right\|^2 \right)^{1/2} \cdot \sup_{\lambda \in \Lambda} \left\| \tilde{\theta}_n^\lambda - \theta_0 \right\| \\ & = n^{-1/2} \cdot O_{\mu_n}(1) \cdot O_{\mu_n}(1) \cdot O_{\mu_n}(1) = O_{\mu_n}(n^{-1/2}). \end{aligned}$$

This proves that the first term in the right hand side of (17) is $o_{\mu_n}(1)$ uniformly over λ . By an analogous argument, the second term in the right hand side of (17) is also $o_{\mu_n}(1)$ uniformly over λ .

Step 3:

It thus remains to show

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \left\langle \tilde{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^\lambda, \sqrt{n} \nabla_{\theta} l_n(\theta_0, Z_n^i) \right\rangle$$

is asymptotically equivalent to the degrees of freedom term in SURE.

By the definition of $R^\lambda(\cdot)$, we may write

$$\tilde{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^\lambda = -\frac{1}{\sqrt{n}} X_n^i - \frac{1}{\sqrt{n}} \nabla g^\lambda \left(\tilde{\theta}_n \right)^\top X_n^i + R^\lambda(\tilde{\theta}_n^{-i} - \tilde{\theta}_n; \tilde{\theta}_n).$$

We can thus expand

$$\begin{aligned} & \frac{1}{\sqrt{n}} \sum_{i=1}^n \left\langle \tilde{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^\lambda, \sqrt{n} \nabla_{\theta} l_n(\theta_0, Z_n^i) \right\rangle \\ &= \frac{1}{n} \sum_{i=1}^n \left\langle -X_n^i, \sqrt{n} \nabla_{\theta} l_n(\theta_0, Z_n^i) \right\rangle + \frac{1}{n} \sum_{i=1}^n \left\langle \nabla g^\lambda \left(\tilde{\theta}_n \right)^\top X_n^i, -\sqrt{n} \nabla_{\theta} l_n(\theta_0, Z_n^i) \right\rangle \\ &+ \frac{1}{\sqrt{n}} \sum_{i=1}^n \left\langle R^\lambda(\tilde{\theta}_n^{-i} - \tilde{\theta}_n; \tilde{\theta}_n), \sqrt{n} \nabla_{\theta} l_n(\theta_0, Z_n^i) \right\rangle. \end{aligned} \quad (18)$$

The first term in (18) is independent of λ and can therefore be neglected.

The second term in (18) is asymptotically equivalent to the degrees of freedom term in SURE. Indeed, since $\tilde{\theta}_n^{-i} - \tilde{\theta}_n = \nabla_{\theta} l_n(\theta_0, Z_n^i)$, we can write

$$\begin{aligned} & \frac{1}{n} \sum_{i=1}^n \left\langle \nabla g^\lambda \left(\tilde{\theta}_n \right)^\top X_n^i, -\sqrt{n} \nabla_{\theta} l_n(\theta_0, Z_n^i) \right\rangle \\ &= \frac{1}{n} \sum_{i=1}^n \left\langle \nabla g^\lambda \left(\tilde{\theta}_n \right)^\top X_n^i, X_n^i \right\rangle \\ &= \text{Tr} \left[\nabla g^\lambda \left(\tilde{\theta}_n \right)^\top \hat{\Sigma}_n \right], \end{aligned}$$

where

$$\hat{\Sigma}_n := \frac{1}{n} \sum_{i=1}^n (X_n^i)(X_n^i)^\top.$$

But by the law of large numbers, which can be applied here due to Assumption 5(ii), $\hat{\Sigma}_n = \Sigma + o_{\mu_n}(1)$. We thus conclude that

$$\frac{1}{n} \sum_{i=1}^n \left\langle \nabla g^\lambda \left(\tilde{\theta}_n \right)^\top X_n^i, -\sqrt{n} \nabla_{\theta} l_n \left(\theta_0, Z_n^i \right) \right\rangle = \text{Tr} \left[\nabla g^\lambda \left(\tilde{\theta}_n \right)^\top \Sigma \right] + o_{\mu_n}(1),$$

uniformly over $\lambda \in \Lambda$.

It remains to show the third term in (18) is negligible, i.e.,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \left\langle R^\lambda(\tilde{\theta}_n^{-i} - \tilde{\theta}_n; \tilde{\theta}_n), \sqrt{n} \nabla_{\theta} l_n \left(\theta_0, Z_n^i \right) \right\rangle = o_{\mu_n}(1).$$

Recall that $\tilde{\theta}_n^{-i} - \tilde{\theta}_n = -X_n^i/\sqrt{n}$. Fix some $a \in (0, 1/2)$ and $C < \infty$, and expand

$$\begin{aligned} \frac{1}{\sqrt{n}} \sum_{i=1}^n \left\langle R^\lambda(\tilde{\theta}_n^{-i} - \tilde{\theta}_n; \tilde{\theta}_n), \sqrt{n} \nabla_{\theta} l_n \left(\theta_0, Z_n^i \right) \right\rangle &= \frac{1}{\sqrt{n}} \sum_{i=1}^n \left\langle R^\lambda(-X_n^i/\sqrt{n}; \tilde{\theta}_n), -X_n^i \right\rangle \\ &= \frac{1}{\sqrt{n}} \sum_{i=1}^n \left\langle R^\lambda(-X_n^i/\sqrt{n}; \tilde{\theta}_n) \cdot \mathbb{I} \{ \|X_n^i\| \geq Cn^a \}, -X_n^i \right\rangle \\ &\quad + \frac{1}{\sqrt{n}} \sum_{i=1}^n \left\langle R^\lambda(-X_n^i/\sqrt{n}; \tilde{\theta}_n) \cdot \mathbb{I} \{ \|X_n^i\| < Cn^a \}, -X_n^i \right\rangle \\ &:= A_{n1}^\lambda + A_{n2}^\lambda. \end{aligned}$$

We analyze the terms A_{n1}^λ and A_{n2}^λ separately. For the term A_{n1}^λ , observe that by (11),

$$\sup_{\lambda \in \Lambda} |A_{n1}^\lambda| \leq \frac{2}{n} \sum_{i=1}^n \|X_n^i\|^2 \cdot \mathbb{I} \{ \|X_n^i\| \geq Cn^a \}.$$

Consequently, under Assumption 5(ii) and the given choice of a ,

$$\mathbb{E}_{\mu_n} [|A_{n1}^\lambda|] \leq \frac{2}{C^2 n^{2a}} \mathbb{E}_{\mu_n} [\|X_n^i\|^4] \rightarrow 0$$

as $n \rightarrow \infty$. Thus, $\sup_{\lambda \in \Lambda} A_{n1}^\lambda = o_{\mu_n}(1)$.

Next, we show $\sup_{\lambda \in \Lambda} A_{n2}^\lambda = o_{\mu_n}(1)$. Due to exchangeability over i , this

follows if we show that

$$\lim_{n \rightarrow \infty} \sqrt{n} \mathbb{E}_{\mu_n} \left[\left| \left\langle R^\lambda(-X_n^i/\sqrt{n}; \tilde{\theta}_n) \cdot \mathbb{I}_{\Gamma_i}, -X_n^i \right\rangle \right| \right] = 0, \quad (19)$$

where we use $\mathbb{I}_{\Gamma_i}$ as a short-hand for $\mathbb{I} \{ \|X_n^i\| < Cn^a \}$. Now, by the Cauchy-Schwarz inequality,

$$\begin{aligned} & \sqrt{n} \mathbb{E}_{\mu_n} \left[\left| \left\langle R^\lambda(-X_n^i/\sqrt{n}; \tilde{\theta}_n) \cdot \mathbb{I}_{\Gamma_i}, -X_n^i \right\rangle \right| \right] \\ & \leq \mathbb{E}_{\mu_n}^{1/2} \left[\sup_{\lambda \in \Lambda} \frac{\left\| R^\lambda(-X_n^i/\sqrt{n}; \tilde{\theta}_n) \right\|^2}{\| -X_n^i/\sqrt{n} \|^2} \cdot \mathbb{I}_{\Gamma_i} \right] \cdot \mathbb{E}_{\mu_n}^{1/2} \left[\|X_n^i\|^4 \right]. \end{aligned}$$

But $\mathbb{E}_{\mu_n}^{1/2} \left[\|X_n^i\|^4 \right] \leq M < \infty$ under Assumption 5(ii), so (19) would follow if we show that

$$\lim_{n \rightarrow \infty} \mathbb{E}_{\mu_n} \left[\sup_{\lambda \in \Lambda} \frac{\left\| R^\lambda(-X_n^i/\sqrt{n}; \tilde{\theta}_n) \right\|^2}{\| -X_n^i/\sqrt{n} \|^2} \cdot \mathbb{I}_{\Gamma_i} \right] = 0. \quad (20)$$

To prove (20), observe that it is without loss of generality to suppose $R^\lambda(\delta; \theta) = R^\lambda(\|\delta\|; \theta)$, i.e., that it depends only on $\|\delta\|$, and that $R^\lambda(\|\delta\|; \theta)/\|\delta\|$ is increasing in $\|\delta\|$. Otherwise, we can simply define

$$\bar{R}^\lambda(\|\delta\|; \theta) = \sup_{\|\delta'\| \leq \|\delta\|} \frac{R^\lambda(\delta'; \theta)}{\|\delta'\|},$$

and this would satisfy these two conditions while still retaining the property (9). Consequently, the left hand side of (20) can be bounded as

$$\begin{aligned} \mathbb{E}_{\mu_n} \left[\sup_{\lambda \in \Lambda} \frac{\left\| R^\lambda(\|X_n^i/\sqrt{n}\|; \tilde{\theta}_n) \right\|^2}{\|X_n^i/\sqrt{n}\|^2} \cdot \mathbb{I}_{\Gamma_i} \right] & \leq \mathbb{E}_{\mu_n} \left[\sup_{\lambda \in \Lambda} \left\| \frac{R^\lambda(Cn^{a-\frac{1}{2}}; \tilde{\theta}_n)}{Cn^{a-\frac{1}{2}}} \right\|^2 \right] \\ & = \mathbb{E}_{\mu_n} \left[\sup_{\lambda \in \Lambda} \left\| B^\lambda(Cn^{a-\frac{1}{2}}, \tilde{\theta}_n) \right\|^2 \right], \end{aligned}$$

where

$$B^\lambda(\delta, \tilde{\theta}_n) := \frac{\|R^\lambda(\delta; \tilde{\theta}_n)\|}{\|\delta\|}.$$

We now bound

$$\mathbb{E}_{\mu_n} \left[\sup_{\lambda \in \Lambda} \|B^\lambda(Cn^{a-\frac{1}{2}}, \tilde{\theta}_n)\|^2 \right].$$

Fix some $\epsilon > 0$. By the requirement that $R^\lambda(\|\delta\|; \theta)/\|\delta\|$ is increasing in $\|\delta\|$, along with the fact $a < 1/2$, there exists $\bar{n}$ large enough so that $B^\lambda(Cn^{a-\frac{1}{2}}, \theta) \leq B^\lambda(\epsilon, \theta)$ for each $n \geq \bar{n}$, $\theta \in \mathbb{R}^d$ and $\lambda \in \Lambda$. Now, it is straightforward to show

$$\tilde{\theta}_n \xrightarrow[\mu_n]{d} Z \sim N(\theta_0, \Sigma).$$

We then have

$$\begin{aligned} & \lim_{n \rightarrow \infty} \mathbb{E}_{\mu_n} \left[\sup_{\lambda \in \Lambda} \|B^\lambda(Cn^{a-\frac{1}{2}}, \tilde{\theta}_n)\|^2 \right] \\ & \leq \lim_{n \rightarrow \infty} \mathbb{E}_{\mu_n} \left[\sup_{\lambda \in \Lambda} \|B^\lambda(\epsilon, \tilde{\theta}_n)\|^2 \right] = \mathbb{E} \left[\sup_{\lambda \in \Lambda} \|B^\lambda(\epsilon, Z)\|^2 \right], \end{aligned}$$

where the equality follows from the properties of weak convergence since equation (11) implies $\sup_{\lambda \in \Lambda} \|B^\lambda(\epsilon, \theta)\| \leq 2$ uniformly over θ . But (10) implies $\lim_{\epsilon' \rightarrow 0} \sup_{\lambda \in \Lambda} B^\lambda(\epsilon', \theta) = 0$ for every $\theta \in \mathbb{R}^d$ excluding a set of Lebesgue measure 0. Since the Gaussian distribution is absolutely continuous with respect to the Lebesgue measure, it then follows by the dominated convergence theorem that

$$\lim_{\epsilon' \rightarrow 0} \mathbb{E} \left[\sup_{\lambda \in \Lambda} \|B^\lambda(\epsilon', Z)\|^2 \right] = 0.$$

Since $\epsilon > 0$ was arbitrary, we conclude

$$\lim_{n \rightarrow \infty} \mathbb{E}_{\mu_n} \left[\sup_{\lambda \in \Lambda} \|B^\lambda(Cn^{a-\frac{1}{2}}, \tilde{\theta}_n)\|^2 \right] = 0.$$

This proves (20).

It remains to prove the claim, made in Step 2, that

$$\sup_{\lambda \in \Lambda} \left(\sum_{i=1}^n \left\| \left(\hat{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^{\lambda, -i} \right) - \left(\hat{\theta}_n^\lambda - \tilde{\theta}_n^\lambda \right) \right\|^2 \right) = o_{\mu_n}(1).$$

For the remainder of this proof, we make a case distinction between Ridge penalties and Lasso penalties.

Step 4 (Ridge):

We first specialize to the case of quadratic penalties. Define $\tilde{L}_n^{-i}(\theta) = \frac{1}{2} \left\| \theta - \tilde{\theta}_n^{-i} \right\|^2$.

Observe that $\tilde{\theta}_n^{\lambda, -i} = \operatorname{argmin}_{\theta} \left\{ \tilde{L}_n^{-i}(\theta) + \lambda \pi(\theta) \right\}$. Consequently,

$$\begin{aligned} & \nabla_{\theta} \left\{ \tilde{L}_n^{-i}(\hat{\theta}_n^{\lambda, -i}) + \lambda \pi(\hat{\theta}_n^{\lambda, -i}) \right\} \\ &= \nabla_{\theta} \left\{ \tilde{L}_n^{-i}(\hat{\theta}_n^{\lambda, -i}) + \lambda \pi(\hat{\theta}_n^{\lambda, -i}) \right\} - \nabla_{\theta} \left\{ \tilde{L}_n^{-i}(\tilde{\theta}_n^{\lambda, -i}) + \lambda \pi(\tilde{\theta}_n^{\lambda, -i}) \right\} \\ &= \left\{ \hat{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^{\lambda, -i} \right\} + \lambda \nabla_{\theta} \left\{ \pi(\hat{\theta}_n^{\lambda, -i}) - \pi(\tilde{\theta}_n^{\lambda, -i}) \right\}. \end{aligned}$$

But $\nabla_{\theta} \left\{ \tilde{L}_n^{-i}(\hat{\theta}_n^{\lambda, -i}) + \lambda \pi(\hat{\theta}_n^{\lambda, -i}) \right\} = 0$, so we obtain

$$\left\{ \hat{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^{\lambda, -i} \right\} + \lambda \nabla_{\theta} \left\{ \pi(\hat{\theta}_n^{\lambda, -i}) - \pi(\tilde{\theta}_n^{\lambda, -i}) \right\} = \nabla_{\theta} \tilde{L}_n^{-i}(\hat{\theta}_n^{\lambda, -i}) - \nabla_{\theta} \tilde{L}_n^{-i}(\tilde{\theta}_n^{\lambda, -i}).$$

In a similar vein,

$$\left\{ \hat{\theta}_n^\lambda - \tilde{\theta}_n^\lambda \right\} + \lambda \nabla_{\theta} \left\{ \pi(\hat{\theta}_n^\lambda) - \pi(\tilde{\theta}_n^\lambda) \right\} = \nabla_{\theta} \tilde{L}_n(\hat{\theta}_n^\lambda) - \nabla_{\theta} \tilde{L}_n(\tilde{\theta}_n^\lambda).$$

When $\pi(\theta)$ is the ridge penalty $\frac{1}{2} \theta^\top A^{-1} \theta$, we have

$$\left(\hat{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^{\lambda, -i} \right) - \left(\hat{\theta}_n^\lambda - \tilde{\theta}_n^\lambda \right) = (I + \lambda A^{-1})^{-1} \left\{ \nabla_{\theta} \left(\tilde{L}_n^{-i} - L_n^{-i} \right) (\hat{\theta}_n^{\lambda, -i}) - \nabla_{\theta} \left(\tilde{L}_n - L_n \right) (\hat{\theta}_n^\lambda) \right\}.$$

By a third order Taylor expansion,

$$\begin{aligned}
& L_n^{-i}(\theta) - L_n^{-i}(\theta_0) \\
&= \frac{1}{\sqrt{n}} \left\{ \sqrt{n} \nabla_{\theta} L_n^{-i}(\theta_0) \right\}^{\top} (\theta - \theta_0) + \frac{1}{2n} (\theta - \theta_0)^{\top} \left\{ n \nabla_{\theta}^2 L_n^{-i}(\theta_0) \right\} (\theta - \theta_0) \\
&\quad + \frac{1}{6n^{3/2}} D_{\theta}^3 L_n^{-i}(\bar{\theta}) [\theta - \theta_0, \theta - \theta_0, \theta - \theta_0],
\end{aligned}$$

for some $\bar{\theta}$ between θ and θ_0 . At the same time,

$$\tilde{L}_n^{-i}(\theta) - \tilde{L}_n^{-i}(\theta_0) = \frac{1}{\sqrt{n}} \left\{ \sqrt{n} \nabla_{\theta} L_n^{-i}(\theta_0) \right\}^{\top} (\theta - \theta_0) + \frac{1}{2} (\theta - \theta_0)^{\top} (\theta - \theta_0).$$

Therefore, by Assumption 5(iv), which implies

$$\mathbb{E}_{\mu_n} \left[\sup_{\theta \in \Theta} \|n^2 D_{\theta}^4 l_n(\theta, Z_n^i)\|^4 \right] \leq M,$$

we obtain

$$\nabla_{\theta} \left(\tilde{L}_n^{-i} - L_n^{-i} \right) (\theta) = \left\{ \nabla_{\theta}^2 L_n^{-i}(\theta_0) - I \right\} (\theta - \theta_0) + O_{\mu_n}(n^{-3/2}).$$

In a similar vein,

$$\nabla_{\theta} \left(\tilde{L}_n - L_n \right) (\theta) = \left\{ \nabla_{\theta}^2 L_n(\theta_0) - I \right\} (\theta - \theta_0) + O_{\mu_n}(n^{-3/2}).$$

Taken together, we conclude

$$\begin{aligned}
& \left\{ \nabla_{\theta} \left(\tilde{L}_n^{-i} - L_n^{-i} \right) (\hat{\theta}_n^{\lambda, -i}) - \nabla_{\theta} \left(\tilde{L}_n - L_n \right) (\hat{\theta}_n^{\lambda}) \right\} \\
&= \left\{ \nabla_{\theta}^2 L_n^{-i}(\theta_0) - I \right\} \left(\hat{\theta}_n^{\lambda, -i} - \theta_0 \right) - \left\{ \nabla_{\theta}^2 L_n(\theta_0) - I \right\} \left(\hat{\theta}_n^{\lambda} - \theta_0 \right) + O_{\mu_n}(n^{-3/2}) \\
&= -\frac{1}{n} \left\{ n \nabla_{\theta}^2 l_n(\theta_0, Z_n^i) \right\} \left(\hat{\theta}_n^{\lambda, -i} - \theta_0 \right) + \left\{ \nabla_{\theta}^2 L_n(\theta_0) - I \right\} \left(\hat{\theta}_n^{\lambda, -i} - \hat{\theta}_n^{\lambda} \right) + O_{\mu_n}(n^{-3/2}).
\end{aligned}$$

Hence,

$$\begin{aligned}
& \sum_{i=1}^n \left\| \left(\hat{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^{\lambda, -i} \right) - \left(\hat{\theta}_n^{\lambda} - \tilde{\theta}_n^{\lambda} \right) \right\|^2 \\
& \leq \frac{2}{n^2} \left(\sum_i \left\| n \nabla_{\theta}^2 l_n(\theta_0, Z_n^i) \right\|^2 \cdot \left\| \hat{\theta}_n^{\lambda, -i} - \theta_0 \right\|^2 \right) \\
& \quad + 2 \left\| \nabla_{\theta}^2 L_n(\theta_0) - I \right\|^2 \cdot \left(\sum_i \left\| \hat{\theta}_n^{\lambda, -i} - \hat{\theta}_n^{\lambda} \right\|^2 \right) + O_{\mu_n}(n^{-3/2}) \\
& = o_{\mu_n}(1).
\end{aligned}$$

Step 4 (Lasso):

We finally prove that for the Lasso penalty, we again have that

$$DD_n := \left(\sum_{i=1}^n \left\| \left(\hat{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^{\lambda, -i} \right) - \left(\hat{\theta}_n^{\lambda} - \tilde{\theta}_n^{\lambda} \right) \right\|^2 \right) = o_{\mu_n}(1).$$

We consider the case of fixed λ ; taking the supremum over λ is trivial when Λ is finite. We will also assume for notational simplicity that $A = I$, so that $h = \theta$; the general case (where $h = A^{-1} \cdot \theta$) follows immediately.

Let now

$$D_n^i := \left(\hat{\theta}_n^{\lambda, -i} - \tilde{\theta}_n^{\lambda, -i} \right) - \left(\hat{\theta}_n^{\lambda} - \tilde{\theta}_n^{\lambda} \right).$$

Denote $\eta = \text{sign}(\tilde{\theta}_n^{\lambda})$, and the set of active coordinates as $J = \{j : \eta_j \neq 0\}$. Define the following three event indicators, where the *sign* function is applied component-wise, and $\rho_n > 0$ for some deterministic sequence ρ_n such that $\rho_n \cdot n^{1/4} \rightarrow \infty$ and $\rho_n \rightarrow 0$:

$$\begin{aligned}
A_n &= \mathbf{1} \left(\text{sign}(\hat{\theta}_n^{\lambda, -i}) = \text{sign}(\tilde{\theta}_n^{\lambda, -i}) = \text{sign}(\hat{\theta}_n^{\lambda}) = \eta \text{ for all } i \right), \\
B_n &= (1 - A_n) \cdot \mathbf{1} \left(\text{sign}(\hat{\theta}_n^{\lambda}) \neq \eta \text{ or } \text{sign}(\tilde{\theta}_n + t + g^{\lambda}(\tilde{\theta}_n + t)) \neq \eta \text{ for some } \|t\| < \rho_n \right) \\
C_n &= 1 - A_n - B_n.
\end{aligned}$$

Then, by construction, $A_n + B_n + C_n = 1$ for all n, i , so that

$$DD_n = \underbrace{A_n \cdot \sum_{i=1}^n \|D_n^i\|^2}_{\text{First sum}} + \underbrace{B_n \cdot \sum_{i=1}^n \|D_n^i\|^2}_{\text{Second sum}} + \underbrace{C_n \cdot \sum_{i=1}^n \|D_n^i\|^2}_{\text{Third sum}}.$$

To show that $DD_n \rightarrow 0$, we will consider each of these three sums separately.

First sum Conditional on $A_n = 1$, the signs for all estimators of θ under consideration coincide with η . The first order conditions for the active coordinates for each of the estimators can therefore be written as

$$\begin{aligned} \nabla_J L_n(\hat{\theta}_n^\lambda) + \lambda \cdot \eta_J &= 0 \\ \nabla_J L_n(\hat{\theta}_n^{\lambda, -i}) - \nabla_J l_n(\hat{\theta}_n^{\lambda, -i}, Z_i^n) + \lambda \cdot \eta_J &= 0 \\ (\tilde{\theta}_n^\lambda - \tilde{\theta}_n)_J + \lambda \cdot \eta_J &= 0 \\ (\tilde{\theta}_n^{\lambda, -i} - \tilde{\theta}_n)_J - \nabla_J l_n(\theta_0, Z_i^n) + \lambda \cdot \eta_J &= 0. \end{aligned}$$

Taking differences of the first two and of the second two equations, we get

$$\begin{aligned} \nabla_J L_n(\hat{\theta}_n^\lambda) - \nabla_J L_n(\hat{\theta}_n^{\lambda, -i}) &= \nabla_J l_n(\hat{\theta}_n^{\lambda, -i}, Z_i^n), \\ (\tilde{\theta}_n^\lambda - \tilde{\theta}_n^{\lambda, -i})_J &= \nabla_J l_n(\theta_0, Z_i^n). \end{aligned}$$

The first of these equations can be rewritten as

$$(\hat{\theta}_n^\lambda - \hat{\theta}_n^{\lambda, -i})_J = (\nabla_J^2 L_n(\bar{\theta}_n^i))^{-1} \cdot \nabla_J l_n(\hat{\theta}_n^{\lambda, -i}, Z_i^n)$$

for some intermediate point $\bar{\theta}^i$. Combining, we get $D_{n,J^c}^i = 0$ (for the inactive coordinates, the double difference vanishes) and

$$\begin{aligned} D_{n,J}^i &= \left[(\nabla_J^2 L_n(\bar{\theta}_n^i))^{-1} \cdot \nabla_J l_n(\hat{\theta}_n^{\lambda, -i}, Z_i^n) \right] - \nabla_J l_n(\theta_0, Z_i^n) \\ &= \left[(\nabla_J^2 L_n(\bar{\theta}_n^i))^{-1} - I_J \right] \cdot \nabla_J l_n(\hat{\theta}_n^{\lambda, -i}, Z_i^n) + \left[\nabla_J l_n(\hat{\theta}_n^{\lambda, -i}, Z_i^n) - \nabla_J l_n(\theta_0, Z_i^n) \right]. \end{aligned}$$

and thus, conditional on $A_n = 1$,

$$\begin{aligned}
\sum_{i=1}^n \|D_n^i\|^2 &= \sum_{i=1}^n \|D_{n,J}^i\|^2 \\
&\leq 2 \max_i \left\| (\nabla_J^2 L_n(\bar{\theta}_n^i))^{-1} - I_J \right\|^2 \cdot \sum_i \left\| \nabla l_n(\hat{\theta}_n^{\lambda,-i}, Z_i^n) \right\|^2 \\
&\quad + 2 \sum_i \left\| \nabla l_n(\hat{\theta}_n^{\lambda,-i}, Z_i^n) - \nabla l_n(\theta_0, Z_i^n) \right\|^2.
\end{aligned}$$

Note that we dropped the J subscript on gradients when taking the upper bound. The max term goes to 0 in probability by Lemma 2. The first sum is $O_p(1)$ given the bound on 4th moments of the score in Assumption 5.2. The second sum is $o_p(1)$ by the Lipschitz condition in Assumption 5.2 and by $\hat{\theta}_n^{\lambda,-i} - \theta_0 = O_p\left(\frac{1}{\sqrt{n}}\right)$. It follows that $A_n \cdot \sum_{i=1}^n \|D_n^i\|^2 = o_p(1)$.

Second sum To control the second sum, we next show that $B_n = o_{\mu_n}(1)$. Recall the following results that were shown previously:

- By Lemma 2, $\hat{\theta}_n^\lambda - \tilde{\theta}_n^\lambda \rightarrow^p 0$.
- Also by Lemma 2, $\sup_{\theta: \|\theta\| \leq C} \nabla \epsilon_n(\theta) = o_{\mu_n}(1)$.
- By Corollary 2, $\hat{\theta}_n^\lambda \rightarrow^d \hat{\theta} + g^\lambda(\hat{\theta})$, where $\hat{\theta} \sim N(\theta_0, \Sigma)$.
- By Lemma 8 below, for all $\delta > 0$ there exists a $\gamma > 0$ such that $g^\lambda(\theta)$ is linear on $S_\gamma(\hat{\theta}) = \{\theta : \|\theta - \hat{\theta}\| < \gamma\}$ with probability greater than $1 - \delta$, where $\hat{\theta} \sim N(\theta_0, \Sigma)$.

We claim that the combination of these results implies that $B_n \rightarrow^p 0$ as long as $\rho_n \rightarrow 0$. To see this, define $\Theta_\gamma = \{\theta : g^\lambda \text{ is linear (affine) on } S_\gamma(\theta)\}$, for $\gamma > 0$. By Lemma 8, $P(\hat{\theta} \in \Theta_\gamma) > 1 - \delta$ for γ small enough. By Corollary 2, and the definition of convergence in distribution, we therefore get $P(\hat{\theta}_n \in \Theta_\gamma) > 1 - \delta$ for n large enough.

By Lemma 2, $\|\hat{\theta}_n^\lambda - \tilde{\theta}_n^\lambda\| < \gamma$ with probability greater than $1 - \delta$ for n large enough. This implies that $\text{sign}(\hat{\theta}_n^\lambda)_J = \eta_J$, because $|\tilde{\theta}_{n,j}^\lambda|$ is bounded away from 0 for $j \in J$, by definition of Θ_γ . This takes care of the active coordinates.

Let now $j \in J^c$ be one of the inactive coordinates. If $\tilde{\theta}_n \in \Theta_\gamma$, then necessarily $|\tilde{\theta}_{j,n}| + \gamma < \lambda$, by definition of Θ_γ , since the mapping from $\tilde{\theta}_{j,n}$ to $\tilde{\theta}_{j,n}^\lambda$ has a kink at $\pm\lambda$.

Let θ, θ' be equal in all coordinates except j , where $\theta_j = t$ and $\theta'_j = 0$. Then, by Lemma 2, by $|\tilde{\theta}_{j,n}| + \gamma < \lambda$, and by the Lipschitz continuity of $\epsilon(\theta)$ with constant $\gamma_n < \gamma$, for n large enough, which follows again from Lemma 2,

$$\begin{aligned} & (L_n(\theta) + \lambda\|\theta\|_1) - (L_n(\theta') + \lambda\|\theta'\|_1) \\ &= \frac{1}{2}(t - \tilde{\theta}_{j,n})^2 + \epsilon(\theta) + \lambda|t| - \frac{1}{2}\tilde{\theta}_{j,n}^2 - \epsilon(\theta') \\ &\geq \frac{1}{2}t^2 + |t| \cdot (-(\lambda - \gamma) + \lambda - \gamma_n) \\ &= \frac{1}{2}t^2 + |t| \cdot (\gamma - \gamma_n) \\ &\geq 0, \end{aligned}$$

with equality only for $t = 0$. It follows that the minimizer of $L_n(\theta) + \lambda\|\theta\|_1$ necessarily has j th component equal to zero. This takes care of the inactive coordinates.

Third sum To control the third sum, we show that $C_n \rightarrow^p 0$. Since $\sum_{i=1}^n \|D_n^i\|^2 = O_p(1)$ (which follows from $\|D_n^i\| \leq \|\hat{\theta}_n^{\lambda,-i} - \tilde{\theta}_n^\lambda\| + \|\tilde{\theta}_n^{\lambda,-i} - \tilde{\theta}_n^\lambda\|$ and the bounds (4)–(5), by the same argument as for (13)), this implies $C_n \cdot \sum_i \|D_n^i\|^2 = o_p(1)$.

Recall that $C_n = 1$ requires: (a) $\text{sign}(\hat{\theta}_n^\lambda) = \eta$ and g^λ is linear on $B_{\rho_n}(\tilde{\theta}_n)$ (the negation of B_n 's condition), but (b) there exists some i such that $\text{sign}(\hat{\theta}_n^{\lambda,-i}) \neq \eta$ or $\text{sign}(\tilde{\theta}_n^{\lambda,-i}) \neq \eta$. We show that on the event described in (a), neither type of sign disagreement occurs, with probability approaching 1.

Preliminary: rate for the influence-function approximation error. We claim

$$\sup_\lambda \|\hat{\theta}_n^\lambda - \tilde{\theta}_n^\lambda\| = O_{\mu_n}(n^{-1/2}) = o_{\mu_n}(\rho_n), \quad (21)$$

where the second equality uses $\rho_n n^{1/2} = (\rho_n n^{1/4}) \cdot n^{1/4} \rightarrow \infty$. By Lemma 2, $\hat{\theta}_n^\lambda$ and $\tilde{\theta}_n^\lambda$ are respectively the minimizers of $\frac{1}{2}\|\theta - \tilde{\theta}_n\|^2 + \epsilon_n(\theta) + \lambda\pi(\theta)$ and $\frac{1}{2}\|\theta - \tilde{\theta}_n\|^2 + \lambda\pi(\theta)$. By Lemma 1 in Wilson et al. (2020) applied to the

perturbation ϵ_n ,

$$\sup_{\lambda} \|\hat{\theta}_n^\lambda - \tilde{\theta}_n^\lambda\| \leq \frac{1}{\mu} \sup_{\|\theta\| \leq C} \|\nabla \epsilon_n(\theta)\|. \quad (22)$$

Since $\nabla L_n(\theta_0) = \theta_0 - \tilde{\theta}_n$ exactly (by the definition $\tilde{\theta}_n = \theta_0 - \sum_i \nabla_{\theta} l_n(\theta_0, Z_n^i)$ in Lemma 2) and $\nabla \epsilon_n(\theta) = \nabla L_n(\theta) - (\theta - \tilde{\theta}_n)$, we have $\nabla \epsilon_n(\theta_0) = 0$ exactly. Therefore, for $\|\theta\| \leq C$,

$$\|\nabla \epsilon_n(\theta)\| = \|\nabla \epsilon_n(\theta) - \nabla \epsilon_n(\theta_0)\| \leq C \cdot \sup_{\|\theta'\| \leq C} \|\nabla^2 \epsilon_n(\theta')\|.$$

Now $\nabla^2 \epsilon_n(\theta) = \nabla^2 L_n(\theta) - I$. The CLT applied to $n \nabla_{\theta}^2 l_n(\theta_0, Z_n^i)$ (finite second moments from Assumption 5(iii)) gives $\|\nabla^2 L_n(\theta_0) - I\| = O_{\mu_n}(n^{-1/2})$, using that $\|E_{\mu_n}[n \nabla_{\theta}^2 l_n(\theta_0, Z_n^i)] - I\| = O(n^{-1/2})$ by Assumption 2 and the Lipschitz condition in Assumption 5(iii). The same Lipschitz condition and the functional CLT extend this rate uniformly over bounded neighborhoods:

$$\sup_{\|\theta\| \leq C} \|\nabla \epsilon_n(\theta)\| = O_{\mu_n}(n^{-1/2}) = o_{\mu_n}(\rho_n). \quad (23)$$

Substituting into (22) establishes (21).

Sign agreement for $\tilde{\theta}_n^{\lambda, -i}$. Since $\tilde{\theta}_n^{-i} - \tilde{\theta}_n = -\frac{1}{\sqrt{n}} X_n^i$ where $X_n^i = -\nabla_{\beta} l(\theta_0/\sqrt{n}, Z_n^i)$, the event $\|\tilde{\theta}_n^{-i} - \tilde{\theta}_n\| > \rho_n$ for some i has probability bounded by

$$\sum_{i=1}^n P\left(\frac{1}{\sqrt{n}} \|X_n^i\| > \rho_n\right) \leq n \cdot \frac{E_{\mu_n}[\|X_n^i\|^4]}{(\rho_n \sqrt{n})^4} = \frac{M}{(\rho_n n^{1/4})^4} \rightarrow 0,$$

by the 4th-moment bound in Assumption 5(ii) and $\rho_n \cdot n^{1/4} \rightarrow \infty$. On the complement of this event, $\tilde{\theta}_n^{-i} \in B_{\rho_n}(\tilde{\theta}_n)$ for every i , so g^λ is linear on a neighborhood of each $\tilde{\theta}_n^{-i}$, and therefore $\text{sign}(\tilde{\theta}_n^{\lambda, -i}) = \eta$ for all i .

Sign agreement for $\hat{\theta}_n^{\lambda, -i}$: active coordinates. On the event $C_n = 1$, g^λ is linear on $B_{\rho_n}(\tilde{\theta}_n)$. For $j \in J$, this forces $|\tilde{\theta}_{n,j}| > \lambda + \rho_n$, hence $|\tilde{\theta}_{n,j}^\lambda| = |\tilde{\theta}_{n,j}| - \lambda > \rho_n$. By (21), $\sup_{\lambda} \|\hat{\theta}_n^\lambda - \tilde{\theta}_n^\lambda\| = o_{\mu_n}(\rho_n)$, so with probability approaching 1,

$$|\hat{\theta}_{n,j}^\lambda| \geq \rho_n/2 \quad \text{for all } j \in J. \quad (24)$$

It remains to show $|\widehat{\theta}_{n,j}^{\lambda,-i} - \widehat{\theta}_{n,j}^\lambda| < \rho_n/4$ for all $j \in J$ and all i simultaneously. By (4), $\sup_\lambda \|\widehat{\theta}_n^{\lambda,-i} - \widehat{\theta}_n^\lambda\| \leq \frac{1}{\mu} \|\nabla_\theta l_n(\widehat{\theta}_n, Z_n^i)\|$. The Lipschitz condition in Assumption 5(ii) gives

$$\|\nabla_\theta l_n(\widehat{\theta}_n, Z_n^i)\| \leq \frac{1}{\sqrt{n}} \|X_n^i\| + \frac{B_n(Z_n^i)}{\sqrt{n}} \|\widehat{\theta}_n - \theta_0\|.$$

A Chebyshev union bound on the first term (4th-moment bound in Assumption 5(ii)) and a Chebyshev union bound on the second term (2nd-moment bound $E_{\mu_n}[B_n^2] < \infty$ in Assumption 5(ii), together with $\widehat{\theta}_n - \theta_0 = O_{\mu_n}(1)$) give

$$P\left(\exists i : \sup_\lambda \|\widehat{\theta}_n^{\lambda,-i} - \widehat{\theta}_n^\lambda\| > \frac{\rho_n}{4}\right) \leq \frac{C_1 M}{(\rho_n n^{1/4})^4} + \frac{C_2 E_{\mu_n}[B_n^2]}{n \rho_n^2} \rightarrow 0, \quad (25)$$

for absolute constants C_1, C_2 , using $(\rho_n n^{1/4})^4 \rightarrow \infty$ and $n \rho_n^2 \geq (\rho_n n^{1/4})^2 \cdot n^{1/2} \rightarrow \infty$. On the complement of (25) and given (24), we have $|\widehat{\theta}_{n,j}^{\lambda,-i}| \geq \rho_n/4 > 0$ with sign η_j , for all $j \in J$ and all i .

Sign agreement for $\widehat{\theta}_n^{\lambda,-i}$: inactive coordinates. It remains to show that $\widehat{\theta}_{n,j}^{\lambda,-i} = 0$ for $j \in J^c$ and all i , with probability approaching 1. The argument parallels the one used for the second sum (showing $\text{sign}(\widehat{\theta}_n^\lambda) = \eta$), now applied to L_n^{-i} and $\widehat{\theta}_n^{\lambda,-i}$.

By Lemma 2,

$$L_n^{-i}(\theta) = \frac{1}{2} \|\theta - \tilde{\theta}_n^{-i}\|^2 + \epsilon_n^{(i)}(\theta) + c_n^{(i)},$$

where

$$\begin{aligned} \epsilon_n^{(i)}(\theta) &= \epsilon_n(\theta) - [l_n(\theta, Z_n^i) - l_n(\theta_0, Z_n^i) - \langle \nabla_\theta l_n(\theta_0, Z_n^i), \theta - \theta_0 \rangle \\ &\quad + \frac{1}{2n} (\theta - \theta_0)^\top \{n \nabla_\theta^2 l_n(\theta_0, Z_n^i)\} (\theta - \theta_0)]. \end{aligned}$$

Consider a candidate minimizer θ of $L_n^{-i}(\theta) + \lambda \|\theta\|_1$ with $\theta_j = t \neq 0$ for some $j \in J^c$, and let θ' agree with θ except $\theta'_j = 0$. On the event that g^λ is linear on $B_{\rho_n}(\tilde{\theta}_n)$, the KKT conditions imply $|\tilde{\theta}_{n,j}| + \rho_n < \lambda$ for $j \in J^c$. Therefore,

by the same calculation as in the second sum,

$$(L_n^{-i}(\theta) + \lambda|t|) - L_n^{-i}(\theta') \geq \frac{1}{2}t^2 + |t| \left(\lambda - |\tilde{\theta}_{n,j}^{-i}| - \sup_{\|\theta\| \leq C} \|\nabla \epsilon_n^{(i)}(\theta)\| \right).$$

Since $|\tilde{\theta}_{n,j}^{-i}| \leq |\tilde{\theta}_{n,j}| + n^{-1/2}\|X_n^i\|$ and the slack is $\lambda - |\tilde{\theta}_{n,j}| > \rho_n$ on the event under consideration, it suffices to show simultaneously for all i :

$$n^{-1/2}\|X_n^i\| + \sup_{\|\theta\| \leq C} \|\nabla \epsilon_n^{(i)}(\theta)\| < \rho_n. \quad (26)$$

Write $\nabla \epsilon_n^{(i)}(\theta) = \nabla \epsilon_n(\theta) - R_3(\theta, Z_n^i)$, where

$$R_3(\theta, Z_n^i) := \nabla_{\theta} l_n(\theta, Z_n^i) - \nabla_{\theta} l_n(\theta_0, Z_n^i) - \frac{1}{n} \{n \nabla_{\theta}^2 l_n(\theta_0, Z_n^i)\}(\theta - \theta_0)$$

is the second-order Taylor remainder of $\nabla_{\theta} l_n(\cdot, Z_n^i)$ around θ_0 . We bound the three contributions to (26) in turn.

Control of $\nabla \epsilon_n$. By (23), $\sup_{\|\theta\| \leq C} \|\nabla \epsilon_n(\theta)\| = O_{\mu_n}(n^{-1/2}) = o_{\mu_n}(\rho_n)$; this term is the same for all i .

Simultaneous control of $R_3(\cdot, Z_n^i)$. By the Lipschitz condition in Assumption 5(iii), $\|R_3(\theta, Z_n^i)\| \leq \frac{C_n(Z_n^i)}{2n} \|\theta - \theta_0\|^2$, so $\sup_{\|\theta\| \leq C} \|R_3(\theta, Z_n^i)\| \leq \frac{C_n(Z_n^i)C^2}{2n}$. A Chebyshev union bound gives

$$P\left(\exists i : \sup_{\|\theta\| \leq C} \|R_3(\theta, Z_n^i)\| > \frac{\rho_n}{3}\right) \leq n \cdot P\left(C_n(Z_n^i) > \frac{2n\rho_n}{3C^2}\right) \leq \frac{9C^4 \sup_n E_{\mu_n}[C_n(Z_n^i)^2]}{4n\rho_n^2} \rightarrow 0, \quad (27)$$

since $n\rho_n^2 \geq (\rho_n n^{1/4})^2 \cdot n^{1/2} \rightarrow \infty$ and $\sup_n E_{\mu_n}[C_n(Z_n^i)^2] < \infty$ by Assumption 5(iii).

Simultaneous control of $n^{-1/2}\|X_n^i\|$. By the same Chebyshev argument as for the first sign-agreement step,

$$P\left(\exists i : n^{-1/2}\|X_n^i\| > \frac{\rho_n}{3}\right) \leq \frac{81M}{(\rho_n n^{1/4})^4} \rightarrow 0. \quad (28)$$

On the complement of the three events above, (26) holds for all i : $n^{-1/2}\|X_n^i\| +$

$\sup_{\|\theta\| \leq C} \|\nabla \epsilon_n^{(i)}(\theta)\| \leq \rho_n/3 + o_{\mu_n}(\rho_n) + \rho_n/3 < \rho_n$ for n large enough. The slack is therefore strictly positive, so $\widehat{\theta}_{n,j}^{\lambda,-i} = 0$ for all $j \in J^c$ and all i .

Conclusion. Combining the three cases above, we conclude that on the event described in condition (a) (which has probability approaching 1 given $B_n \rightarrow^p 0$), all signs agree: $\text{sign}(\widehat{\theta}_n^{\lambda,-i}) = \text{sign}(\tilde{\theta}_n^{\lambda,-i}) = \eta$ for every i . This contradicts condition (b), so $C_n \rightarrow^p 0$. Since $\sum_i \|D_n^i\|^2 = O_p(1)$, we obtain $C_n \cdot \sum_i \|D_n^i\|^2 = o_p(1)$.

Combining the bounds for all three sums, we conclude $DD_n = o_{\mu_n}(1)$, which establishes the claim deferred from Step 2.

Conclusion of the proof. It remains to assemble the pieces. By Steps 1–3, uniformly over $\lambda \in \Lambda$,

$$CV_n(\lambda) = \sum_{i=1}^n l_n(\hat{\theta}_n^\lambda, Z_n^i) + \text{Tr} \left[\nabla g^\lambda(\tilde{\theta}_n) \cdot \Sigma \right] + c'_n + o_{\mu_n}(1),$$

where c'_n collects the λ -independent terms isolated in Step 3. For the in-sample term, Lemma 2—applied at $\theta = \hat{\theta}_n^\lambda$, using $\sup_\lambda \|\hat{\theta}_n^\lambda - \tilde{\theta}_n^\lambda\| = o_{\mu_n}(1)$ with $\tilde{\theta}_n^\lambda = \tilde{\theta}_n + g^\lambda(\tilde{\theta}_n)$, and the Lipschitz property of g^λ (Lemma 1)—gives

$$\sum_{i=1}^n l_n(\hat{\theta}_n^\lambda, Z_n^i) = L_n(\hat{\theta}_n^\lambda) = c''_n + \frac{1}{2} \|g^\lambda(\hat{\theta}_n)\|^2 + o_{\mu_n}(1),$$

uniformly over λ , where $c''_n = L_n(\theta_0) - \frac{1}{2} \|\tilde{\theta}_n - \theta_0\|^2$ is λ -independent. Since $\hat{\Sigma}_n \rightarrow^{\mu_n} \Sigma$ and $\nabla g^\lambda(\tilde{\theta}_n) = \nabla g^\lambda(\hat{\theta}_n) + o_{\mu_n}(1)$ (from $\hat{\theta}_n = \tilde{\theta}_n + o_{\mu_n}(1)$ and Lemma 8), collecting c'_n , c''_n , and the λ -independent $\frac{1}{2} \text{trace}(\Sigma)$ into a single constant c_n yields

$$CV_n(\lambda) = c_n + \frac{1}{2} \|g^\lambda(\hat{\theta}_n)\|^2 + \text{Tr} \left[\nabla g^\lambda(\hat{\theta}_n) \cdot \Sigma \right] + o_{\mu_n}(1) = c_n + \text{SURE}(\lambda, \hat{\theta}_n, \Sigma) + o_{\mu_n}(1),$$

uniformly over $\lambda \in \Lambda$, where the last equality uses the definition $\text{SURE}(\lambda, \hat{\theta}, \Sigma) = \frac{1}{2} [\text{trace}(\Sigma) + \|g^\lambda(\hat{\theta})\|^2 + 2 \text{Tr}(\nabla g^\lambda(\hat{\theta}) \cdot \Sigma)]$. This completes the proof of Lemma 4.

□

D Characterizing SURE for Ridge

To prove Lemma 5 for the Ridge penalty $\frac{1}{2}\theta \cdot A^{-1} \cdot \theta$, we start by deriving a series of properties of the function $SURE(\lambda, \hat{\theta}, \Sigma) = \frac{1}{2} [\text{trace}(\Sigma) + \|g^\lambda\|^2 + 2 \text{trace}(\nabla g^\lambda \cdot \Sigma)]$. The properties derived in this section are purely analytic, not probabilistic, and concern the behavior of $SURE$ as a function, not any asymptotic limits. Recall that $g^\lambda(\theta) = \text{argmin}_g \frac{1}{2}\|g\|^2 + \lambda \cdot \pi(\theta + g)$, and that g^λ satisfies the first-order condition,

$$g^\lambda(\theta) = -\lambda \cdot \nabla \pi(\theta + g^\lambda(\theta)),$$

for a suitable sub-gradient $\nabla \pi$ of π . Ridge corresponds to penalties of the form $\pi(\theta) = \frac{1}{2}\theta \cdot A^{-1} \cdot \theta$, where A is positive definite. Denote $C_\lambda = -(\frac{1}{\lambda}A + I)^{-1}$. The first order condition for $g^\lambda(\theta)$ then implies

$$g^\lambda(\theta) = C_\lambda \cdot \theta, \quad \nabla g^\lambda(\theta) = C_\lambda.$$

and thus

$$SURE(\lambda, R, \nu) = \frac{1}{2} [\text{trace}(\Sigma) + \|C_\lambda \cdot \theta\|^2 + 2 \text{trace}(C_\lambda \cdot \Sigma)].$$

The following change of coordinates will be convenient for some of our arguments. Denote $R_n = \|\hat{\theta}_n\|$ and $\nu_n = \hat{\theta}_n/R_n$, and similarly for R and ν . In a slight abuse of notation, we shall write

$$SURE(\lambda, R, \nu) = SURE(\lambda, \hat{\theta}, \Sigma).$$

The following lemma characterizes the behavior of $SURE$ for Ridge. Property 1 and supermodularity are derived directly from the expression for $SURE$. Properties 2a, 2b, and 3 are then consequences of supermodularity.

Lemma 6 (Properties of $SURE$ for Ridge). *Suppose that $\pi(\theta) = \frac{1}{2}\theta \cdot A^{-1} \cdot \theta$, where A is positive definite. Then the following holds:*

1. For every point θ , the function $SURE(\lambda, \theta)$ satisfies

$$\sup_{\lambda \in \Lambda} |SURE(\lambda, \theta') - SURE(\lambda, \theta)| \rightarrow 0$$

as $\theta' \rightarrow \theta$.

2. The function $SURE(\lambda, R, \nu)$ is strictly supermodular in λ and R . This implies:

(a) $\lambda(R, \nu) = \operatorname{argmin}_{\lambda \in \mathbb{R}^+} SURE(\lambda, R, \nu)$ is monotonically decreasing in R , given ν .

(b) $\lambda(R, \nu)$ has at most countably many discontinuities, as a function of R , given ν .

3. Fix ν and R such that $\lambda(\cdot)$ is continuous in R at (R, ν) , and let $\bar{\lambda} = \lambda(R, \nu)$. Then supermodularity implies that the minimum of $SURE$ is well separated: For any $\epsilon > 0$,

$$\inf_{\lambda \in \mathbb{R}^+ \setminus [\bar{\lambda} - \epsilon, \bar{\lambda} + \epsilon]} SURE(\lambda, R, \nu) - SURE(\bar{\lambda}, R, \nu) > 0.$$

Proof of Lemma 6 (Properties of SURE for Ridge):

1. We have

$$\begin{aligned} & 2 [SURE(\lambda, \theta') - SURE(\lambda, \theta)] \\ &= \|C_\lambda \cdot \theta'\|^2 - \|C_\lambda \cdot \theta\|^2 \\ &\leq \|C_\lambda\|^2 \cdot \|\theta' - \theta\| \cdot \|\theta' + \theta\| \\ &\leq \|\theta' - \theta\| \cdot \|\theta' + \theta\|, \end{aligned}$$

where in the third line we have used Cauchy-Schwartz ($\|\theta'\|^2 - \|\theta\|^2 = \langle \theta' - \theta, \theta' + \theta \rangle \leq \|\theta' - \theta\| \cdot \|\theta' + \theta\|$), and in the last line we have used that A is positive definite, which implies $\|C_\lambda\| \leq 1$. The claim follows.

2. The first and last terms in the expression for $SURE$ do not depend on θ . The middle term can be written as $\frac{1}{2}R^2 \cdot \|C_\lambda \cdot \nu\|^2$, and thus

$$\frac{\partial^2}{\partial \lambda \partial R} SURE(\lambda, R, \nu) = R \cdot \frac{\partial}{\partial \lambda} \|C_\lambda \cdot \nu\|^2 > 0,$$

using again the positive definiteness of A . This implies that $SURE(\lambda, R, \nu)$ is strictly supermodular in λ and R , i.e., whenever $\epsilon > 0$ and $\delta > 0$

$$\begin{aligned} & [SURE(\lambda + \epsilon, R, \nu) - SURE(\lambda, R, \nu)] \\ & - [SURE(\lambda + \epsilon, R - \delta, \nu) - SURE(\lambda, R - \delta, \nu)] > 0. \end{aligned}$$

- (a) Monotonicity of $\lambda(R, \nu)$ in R follows from supermodularity of $SURE$, by Topkis's theorem.
 - (b) That $\lambda(R, \nu)$ is continuous in R almost everywhere holds because monotonic functions are continuous almost everywhere: The set of discontinuity points is at most countable, by Theorem 4.30 of Rudin (1991).
3. For the given R, ν , let δ be such that $|\lambda(R', \nu) - \lambda(R, \nu)| < \epsilon/2$ whenever $|R' - R| \leq \delta$; such a δ exists by continuity.

Define

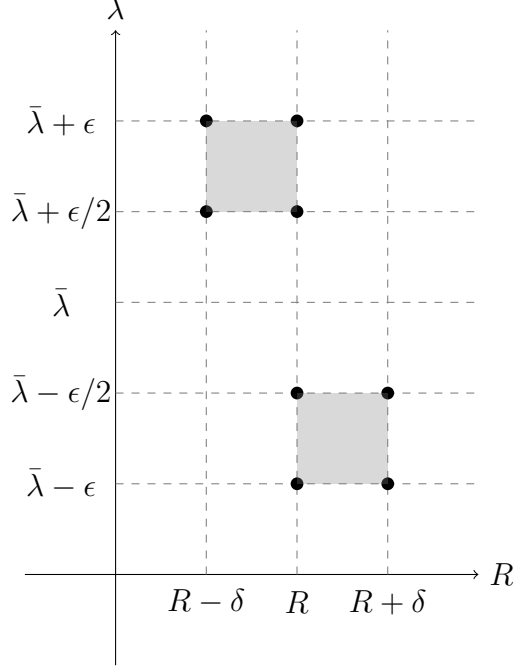
$$A(\lambda_1, \lambda_2, R) = SURE(\lambda_1, R, \nu) - SURE(\lambda_2, R, \nu),$$

let $\bar{\lambda} = \lambda(R, \nu)$ and

$$\begin{aligned} \bar{\Delta} = \min & \left(A(\bar{\lambda} + \epsilon, \bar{\lambda} + \epsilon/2, R) - A(\bar{\lambda} + \epsilon, \bar{\lambda} + \epsilon/2, R - \delta), \right. \\ & \left. A(\bar{\lambda} - \epsilon/2, \bar{\lambda} - \epsilon, R + \delta) - A(\bar{\lambda} - \epsilon/2, \bar{\lambda} - \epsilon, R) \right). \end{aligned}$$

By strict supermodularity, both of the “double differences” in this definition are positive, and thus $\bar{\Delta} > 0$. The following figure illustrates the definition of $\bar{\Delta}$. The differences defining A are taken over vertical segments for different λ and fixed R . The double differences defining Δ

are taken over of the grey rectangles in the figure:



We claim that

$$\inf_{\lambda \in \mathbb{R}^+ \setminus [\bar{\lambda} - \epsilon, \bar{\lambda} + \epsilon]} SURE(\lambda, R, \nu) - SURE(\bar{\lambda}, R, \nu) \geq \bar{\Delta}.$$

The following argument will correspond to the “top left” rectangle in the figure; the “bottom right” case is symmetric. Fix $\tilde{\lambda} > \bar{\lambda} + \epsilon$.

- Given our assumptions and definitions, $\tilde{\lambda} > \bar{\lambda} + \epsilon > \bar{\lambda} + \epsilon/2 > \lambda(R - \delta) \geq \bar{\lambda}$. Therefore, by supermodularity

$$\begin{aligned} & A(\tilde{\lambda}, \lambda(R - \delta), R) - A(\tilde{\lambda}, \lambda(R - \delta), R - \delta) \\ & \geq A(\bar{\lambda} + \epsilon, \bar{\lambda} + \epsilon/2, R) - A(\bar{\lambda} + \epsilon, \bar{\lambda} + \epsilon/2, R - \delta). \end{aligned}$$

- By definition of $\bar{\Delta}$,

$$A(\bar{\lambda} + \epsilon, \bar{\lambda} + \epsilon/2, R) - A(\bar{\lambda} + \epsilon, \bar{\lambda} + \epsilon/2, R - \delta) \geq \bar{\Delta}.$$

- By optimality of $\lambda(R - \delta)$ for $R - \delta$,

$$A(\tilde{\lambda}, \lambda(R - \delta), R - \delta) \geq 0.$$

- Combining the preceding three items, we get

$$A(\tilde{\lambda}, \lambda(R - \delta), R) \geq \bar{\Delta}.$$

and thus

$$SURE(\tilde{\lambda}, R, \nu) \geq SURE(\lambda(R - \delta), R, \nu) + \bar{\Delta} \geq SURE(\bar{\lambda}, R, \nu) + \bar{\Delta}.$$

The argument for $\tilde{\lambda} < \bar{\lambda} - \epsilon$ is analogous, and the claim follows.

□

E Characterizing SURE for Lasso

Lasso corresponds to penalties of the form $\pi(\theta) = \|A^{-1} \cdot \theta\|_1$, where A is an invertible matrix, and $\|\cdot\|_1$ is the L_1 norm.⁸ Given λ , we can characterize $g^\lambda(\theta)$ as follows. Denote $h^\lambda(\theta) = A^{-1}(\theta + g^\lambda(\theta))$. The optimal $h^\lambda(\theta)$ solves

$$h^\lambda(\theta) = \underset{h}{\operatorname{argmin}} \frac{1}{2} \|A \cdot h - \theta\|^2 + \lambda \cdot \|h\|_1.$$

The solution to this convex optimization problem is of the form

$$h_J^\lambda(\theta) = (A_J' A_J)^{-1} \cdot [A_J' \theta - \lambda \eta_J].$$

where $\eta_j = \operatorname{sign}(h_j^\lambda)$, $J = \{j : \eta_j \neq 0\}$, and A_J is the subset of columns corresponding to the index set J . These conditions follow immediately from the first order conditions for $h^\lambda(\theta)$.

⁸For Lasso, it is *not* without loss of generality to assume that A is diagonal.

Lemma 7 (Properties of *SURE* for Lasso). *Suppose that $\pi(\theta) = \|A^{-1} \cdot \theta\|_1$, where A is an invertible matrix, and $\|\cdot\|_1$ is the L_1 norm. Let $k = \dim(\theta)$. Then the following holds:*

1. *As a function of λ , for every R, ν , the graph of $SURE(\lambda, R, \nu)$ consists of at most 3^k continuous segments indexed by $\eta \in \{-1, 0, 1\}^k$. On each of these segments η and $J = \{j : \eta_j \neq 0\}$ are constant,*

$$\begin{aligned}\nabla g^\lambda &= A_J \cdot (A_J' A_J)^{-1} \cdot A_J' - I, \\ \|g^\lambda\|^2 &= \|\nabla g^\lambda \cdot \theta\|^2 + \lambda^2 \cdot \eta_J' (A_J' A_J)^{-1} \eta_J,\end{aligned}$$

and $SURE(\lambda, R, \nu)$ is a monotonically increasing quadratic polynomial in λ of the form

$$SURE(\lambda, R, \nu) = \text{const.} + \frac{1}{2} \lambda^2 \cdot \eta_J' (A_J' A_J)^{-1} \eta_J.$$

2. *$SURE$ for Lasso scales with R as follows:*

$$SURE(R \cdot \lambda, R, \nu) = \frac{1}{2} [\text{trace}(\Sigma) + R^2 \cdot \|g^\lambda(\nu)\|^2 + 2 \text{trace}(\nabla g^\lambda(\nu) \cdot \Sigma)].$$

Given ν , let $\lambda_1, \lambda_2, \dots, \lambda_m$ ($m \leq 3^k$) be the local minimizers of $SURE(\lambda, 1, \nu)$, corresponding to values of λ where η changes. The local minimizers of $SURE(\lambda, R, \nu)$ are then given by $R \cdot \lambda_1, R \cdot \lambda_2, \dots, R \cdot \lambda_m$.

3. *Let $\lambda(R, \nu) = \arg\min_{\lambda \in \mathbb{R}^+} SURE(\lambda, R, \nu)$. Then, given ν , $\lambda(R, \nu) = R \cdot \lambda_{j(R)}$, where $j(R)$ is a monotonically decreasing mapping from R to $1, 2, \dots, m$, and $\lambda_1, \lambda_2, \dots, \lambda_m$ are as before. The graph $\lambda^*(R, \nu)$ thus follows a piecewise linear “sawtooth” pattern with at most 3^k jumps.*
4. *Fix ν and R such that $\lambda(\cdot)$ is continuous in R at (R, ν) , and let $\bar{\lambda} = \lambda(R, \nu)$ be such that $\eta \neq 0$. Then the minimum of $SURE$ is well separated: For any $\epsilon > 0$,*

$$\inf_{\lambda \in \mathbb{R}^+ \setminus [\bar{\lambda} - \epsilon, \bar{\lambda} + \epsilon]} SURE(\lambda, R, \nu) - SURE(\bar{\lambda}, R, \nu) > 0.$$

Proof of Lemma 7 (Properties of SURE for Lasso):

1. Consider two values λ_1, λ_2 of λ such that η is the same for these two values. It follows from the optimality conditions for h^λ that for any intermediate value of λ between λ_1, λ_2 , the optimal h^λ is a linear interpolation between h^{λ_1} and h^{λ_2} , and in particular η remains the same in between λ_1, λ_2 . For details, see Mairal and Yu (2012), Lemma 2.

The vector $\eta \in \{-1, 0, 1\}^k$ can take 3^k possible values. The gradient of g^λ with respect to θ is a function of J , which is a function of η , but it does not depend on λ otherwise:

$$\nabla g^\lambda = A \cdot \nabla h^\lambda - I = A_J \cdot (A_J' A_J)^{-1} \cdot A_J' - I.$$

It follows that the penalty term $2 \text{trace}(\nabla g^\lambda \cdot \Sigma)$ in the expression for *SURE* has at most $3^k - 1$ jumps, as a function of λ for fixed θ , and is constant in between these jumps.⁹

Consider now the term $\|g^\lambda\|^2$, which is the other term in the expression for *SURE*. Fixing λ , and the corresponding set J of active coordinates, we get that $\|g^\lambda\|^2 = \|A_J h_J^\lambda - \theta\|^2$, which is minimized at $\lambda = 0$, holding J fixed. For $\lambda = 0$ we get

$$\|A_J h_J^0 - \theta\|^2 = \|A_J((A_J' A_J)^{-1} A_J' - I) \cdot \theta\|^2 = \|\nabla g^\lambda \cdot \theta\|^2.$$

This is the sum of squared errors for an OLS regression of the elements of θ on the rows of A_J .

Given J , $\|g^\lambda\|^2$ is a quadratic function of λ with second derivative $\partial_\lambda^2 \|A_J h_J^\lambda - \theta\|^2 = 2 \cdot \eta_J' (A_J' A_J)^{-1} \eta_J$. It follows that

$$\|g^\lambda\|^2 = \|\nabla g^\lambda \cdot \theta\|^2 + \lambda^2 \cdot \eta_J' (A_J' A_J)^{-1} \eta_J.$$

This last result implies that $\|g^\lambda\|^2$ is monotonically increasing in λ on each segment defined by η . Since g^λ is continuous in λ (cf. Lemma 2

⁹This bound can be refined, cf. Mairal and Yu (2012), but it is enough for our purposes.

in Mairal and Yu 2012), this also implies that $\|g^\lambda\|^2$ is monotonically increasing across $\lambda \in \mathbb{R}^+$; we will use this fact below.

2. Multiplying the objective of the optimization problem $h^\lambda(\theta) = \operatorname{argmin}_h \frac{1}{2} \|A \cdot h - \theta\|^2 + \lambda \cdot \|h\|_1$ by a factor $1/R^2$ yields

$$h^\lambda(R \cdot \nu) = \operatorname{argmin}_h \frac{1}{2} \|A \cdot (h/R) - \nu\|^2 + \lambda/R \cdot \|h/R\|_1 = R \cdot h^{\lambda/R}(\nu),$$

and thus also

$$\begin{aligned} g^\lambda(R \cdot \nu) &= R \cdot g^{\lambda/R}(\nu), \text{ and} \\ \nabla g^\lambda(R \cdot \nu) &= \frac{1}{R} \cdot \partial_\nu g^\lambda(R \cdot \nu) = \nabla g^{\lambda/R}(\nu). \end{aligned}$$

which immediately implies

$$SURE(R \cdot \lambda, R, \nu) = \frac{1}{2} [\operatorname{trace}(\Sigma) + R^2 \cdot \|g^\lambda(\nu)\|^2 + 2 \operatorname{trace}(\nabla g^\lambda(\nu) \cdot \Sigma)].$$

Turning to the characterization of local minima, since $\|g^\lambda(\nu)\|^2$ is monotonically increasing in λ (cf. item 1), the local minima of $SURE(R \cdot \lambda, R, \nu)$ are exactly the values of λ where $2 \operatorname{trace}(\nabla g^\lambda(\nu) \cdot \Sigma)$ jumps down. These values are independent of R , and the claim follows.

3. That $\lambda(R, \nu) = R \cdot \lambda_{j(R)}$ follows immediately from the preceding item. It remains to show that $j(R)$ is monotonically decreasing. To see this, consider any pair of values $j > j'$. Then $\|g^{\lambda_j}(\nu)\|^2 > \|g^{\lambda_{j'}}(\nu)\|^2$ by monotonicity of $\|g^\lambda(\nu)\|^2$ in λ (cf. item 1), and we get that

$$SURE(R \cdot \lambda_j, R, \nu) - SURE(R \cdot \lambda_{j'}, R, \nu) = \frac{1}{2} R^2 \cdot (\|g^{\lambda_j}(\nu)\|^2 - \|g^{\lambda_{j'}}(\nu)\|^2)$$

is increasing in R . The claim follows.

4. By the preceding argument, at a point of continuity in R

$$\inf_{\lambda \in \{R \cdot \lambda_1, R \cdot \lambda_2, \dots, R \cdot \lambda_m\} \setminus \bar{\lambda}} SURE(\lambda, R, \nu) - SURE(\bar{\lambda}, R, \nu) > 0.$$

The same holds for λ to the right of any of the local minimizers $\{R \cdot \lambda_1, R \cdot \lambda_2, \dots, R \cdot \lambda_m\} \setminus \bar{\lambda}$, since $SURE$ is monotonically increasing in λ away from the local minimizers.

It only remains to verify the condition for λ immediately to the right of the global minimizer $\bar{\lambda}$. This holds because $\|g^\lambda\|^2 = \text{const.} + \lambda^2 \cdot \eta'_J(A'_J A_J)^{-1} \eta_J$ is strictly monotonically increasing in λ for $\lambda > 0$ and $\eta \neq 0$.

□

Lemma 8 (Local linearity of g^λ).

1. For all $\gamma > 0$ there exists an $\epsilon > 0$ such that $g^\lambda(\theta)$ is linear on $B_\epsilon(\hat{\theta}) = \{\theta : \|\theta - \hat{\theta}\| < \epsilon\}$ with probability greater than $1 - \gamma$, where $\hat{\theta} \sim N(\theta_0, \Sigma)$.
2. For almost every point θ , the function $SURE(\lambda, \theta)$ satisfies

$$\sup_{\lambda \in \Lambda} |SURE(\lambda, \theta') - SURE(\lambda, \theta)| \rightarrow 0$$

as $\theta' \rightarrow \theta$.

Proof of Lemma 8: To show the first claim, denote

$$\Theta_\eta = \{\theta : \text{sign}(h^\lambda(\theta)) = \eta\}.$$

By the KKT conditions for h^λ , the set Θ_η is convex for every $\eta \in \{-1, 0, 1\}^k$, and its boundary $\partial\Theta_\eta$ is a finite union of subsets of hyperplanes. Furthermore $\mathbb{R}^k = \bigcup_\eta \Theta_\eta$, and $h^\lambda(\theta)$ is linear in θ on each of the sets Θ_η .

The claim of the lemma therefore follows if we can show that the probability of an ϵ -band around the boundary $\partial\Theta_\eta$,

$$\partial\Theta_\eta^\epsilon = \{\theta : d(\theta, \partial\Theta_\eta) < \epsilon\},$$

where d is the Euclidean distance, has vanishing measure for small ϵ . Because $\partial\Theta_\eta$ is a subset of a finite union of hyperplanes, $P(\hat{\theta} \in \partial\Theta_\eta) = 0$. By the

properties of probability measures, since

$$\partial\Theta_\eta = \bigcap_{\epsilon>0} \partial\Theta_\eta^\epsilon$$

we get

$$0 = P(\hat{\theta} \in \partial\Theta_\eta) = \lim_{\epsilon \rightarrow 0} P(\hat{\theta} \in \partial\Theta_\eta^\epsilon).$$

Therefore, for ϵ small enough, $\hat{\theta}$ is more than ϵ away from the boundary of any of the sets Θ_η with probability bigger than $1 - \gamma$, and the claim follows.

We now turn to the second claim. Fix $\lambda \in \Lambda$. By the preceding argument, for almost all points θ , θ is in the interior of Θ_η , for some η . Thus $\theta' \in \Theta_\eta$, as well, for $\|\theta' - \theta\|$ small enough. By the characterization of *SURE* for Lasso in item 1 of Lemma 7, given λ and η , ∇g^λ is constant and $SURE = \text{const.} + \frac{1}{2}\|g^\lambda\|^2 = \text{const.} + \frac{1}{2}\|\nabla g^\lambda \cdot \theta\|^2$. This is continuous in θ , and thus $|SURE(\lambda, \theta') - SURE(\lambda, \theta)| \rightarrow 0$ as $\theta' \rightarrow \theta$. Since almost everywhere continuity of *SURE* in θ thus holds for fixed λ , it also holds simultaneously for any finite set of λ , and the claim follows. $\square$

F Convergence of risk

The remainder of our proof will draw on some standard results, which we recall here, including the following results from van der Vaart (2000):

1. Joint convergence (item (v) of Theorem 2.6)

If $W_n^1 \rightarrow^d W^1$, and $W_n^2 \rightarrow^p 0$, then $W_n \rightarrow^d W$,
where $W_n = (W_n^1, W_n^2)$, $W = (W^1, 0)$.

2. Almost everywhere CMT (item (ii) of Theorem 2.3): Suppose that $W_n \rightarrow^d W$ (converges in distribution), and that $s(W)$ is almost everywhere continuous. Then $s(W_n) \rightarrow^d s(W)$.

3. Uniform integrability and convergence of expectations (Theo-

rem 2.20): Suppose that $s(W_n) \rightarrow^d s(W)$, and that

$$\lim_{M \rightarrow \infty} \limsup_{n \rightarrow \infty} E [|s(W_n)| \mathbf{1}(|s(W_n)| > M)] = 0. \quad (29)$$

Then $E[s(W_n)] \rightarrow E[s(W)]$.

F.1 Convergence in distribution

Consider some arbitrary function $c(\lambda)$ that is minimized to choose the tuning parameter λ (in due time, we will substitute CV_n for this function). Define

$$\Delta(\lambda) = c(\lambda) - SURE(\lambda, \theta, \Sigma).$$

We think of Δ as an element of the space of bounded functions on $\Lambda \subset \mathbb{R}^+$, endowed with the sup norm. Define furthermore

$$w = (\theta, \Delta), \quad \|w\| = \|\theta\| + \sup_{\lambda \in \Lambda} |\Delta(\lambda)|$$

We have proven for Ridge (in Lemma 6), and for Lasso (in Lemma 7), that the minimum of $SURE$ with respect to λ is well separated for almost all θ (with the exception of Lasso when λ is so large that $\hat{\theta}^\lambda = 0$). This implies the following lemma. The “min” in the definition of $\tilde{\lambda}$ serves as a tie-breaking rule in the case of non-uniqueness of the minimizer.

Lemma 9. *For almost every θ , the mapping from $w = (\theta, \Delta)$ to $g^{\tilde{\lambda}(\theta, \Delta)}$, where*

$$\tilde{\lambda}(\theta, \Delta) = \min \left(\operatorname{argmin}_{\lambda \in \Lambda} [SURE(\lambda, \theta, \Sigma) + \Delta(\lambda)] \right),$$

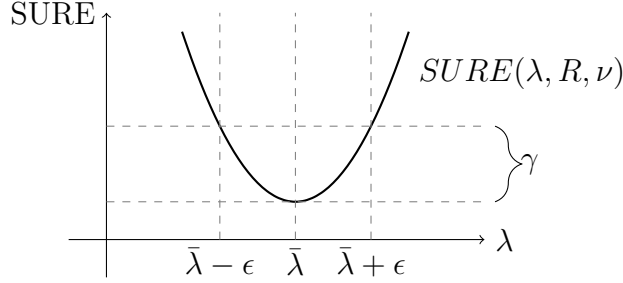
is continuous at $w = (\theta, 0)$ with respect to the norm $\|w\|$.

Proof. We first prove the claim for Ridge, before discussing the necessary modifications for the argument to apply to Lasso. Fix ν . By item 2a of Lemma 6, for almost every R , $\lambda(R, \nu)$ is continuous in R . Fix such an R , and let $\theta = R\nu$.

Fix $\epsilon > 0$ and let

$$\gamma = \inf_{\lambda \in \mathbb{R}^+ \setminus [\bar{\lambda} - \epsilon, \bar{\lambda} + \epsilon]} SURE(\lambda, R, \nu) - SURE(\bar{\lambda}, R, \nu) > 0,$$

where $\bar{\lambda} = \tilde{\lambda}(\theta, 0) \in \operatorname{argmin}_{\lambda} SURE(\lambda, R, \nu)$. The following figure illustrates the definition of γ :



By item 3 of Lemma 6, $\gamma > 0$. By item 1 of Lemma 6, there exists a δ such that if $\|\theta' - \theta\| < \delta$ then $\sup_{\lambda \in \mathbb{R}^+} |SURE(\lambda, \theta') - SURE(\lambda, \theta)| < \gamma/4$.

Let $w' = (\theta', \Delta)$ and $w = (\theta, 0)$. Suppose that $\|w' - w\| < \min(\delta, \gamma/4)$, so that $\|\theta' - \theta\| < \delta$ and $\sup_{\lambda} |\Delta(\lambda)| < \gamma/4$. Denote $c(\lambda) = SURE(\lambda, \theta') + \Delta(\lambda)$, so that $\tilde{\lambda}(\theta', \Delta)$ is a minimizer of $c(\lambda)$. Then

$$\begin{aligned} & SURE(\tilde{\lambda}(\theta', \Delta), \theta) \\ & < SURE(\tilde{\lambda}(\theta', \Delta), \theta') + \frac{1}{4}\gamma \\ & < c(\tilde{\lambda}(\theta', \Delta)) + \frac{1}{2}\gamma \\ & \leq c(\bar{\lambda}) + \frac{1}{2}\gamma \\ & < SURE(\bar{\lambda}, \theta') + \frac{3}{4}\gamma \\ & < SURE(\bar{\lambda}, \theta) + \gamma. \end{aligned}$$

It follows that $|\tilde{\lambda}(\theta', \Delta) - \bar{\lambda}| < \epsilon$. This proves that $\tilde{\lambda}(\theta, \Delta)$ is continuous at $(\theta, 0)$. The claim for Ridge follows, since continuity of $g^\lambda(\theta) = (\frac{1}{\lambda}A + I)^{-1} \cdot \theta$ in both λ and θ is immediate.

Turning to Lasso, most of this argument holds verbatim, with the following modifications: Consider first values of θ such that $\hat{\theta}^{\tilde{\lambda}} \neq 0$.

1. By item 4 of Lemma 7, $\gamma > 0$ for almost all such θ .
2. By item 2 of Lemma 8, for *almost* all θ there exists a δ such that if $\|\theta' - \theta\| < \delta$ then $\sup_{\lambda \in \Lambda} |SURE(\lambda, \theta') - SURE(\lambda, \theta)| < \gamma/4$.
3. By the characterization of g^λ and h^λ given at the outset of Appendix E, $g^\lambda(\theta)$ is continuous in both λ and θ . The claim thus follows for θ such that $\hat{\theta}^{\tilde{\lambda}} \neq 0$.

Consider now values of θ such that $\hat{\theta}^{\tilde{\lambda}} = 0$. For such values, $SURE$ is flat in λ for values of λ greater than $\tilde{\lambda}$, because $\hat{\theta}^\lambda = 0$ and $\nabla g^\lambda = 0$ for all such λ (see Figure 2 for an example). Because $SURE$ is flat to the right, continuity of $\tilde{\lambda}(\theta, \Delta)$ does not necessarily hold at $(\theta, 0)$; small perturbations of Δ can lead to large changes of $\tilde{\lambda}$.

By the same arguments used to prove item 4 of Lemma 7, we obtain however (for almost all such θ) that

$$\inf_{\lambda < \lambda_m} SURE(\lambda, R, \nu) - SURE(\bar{\lambda}, R, \nu) > 0,$$

where λ_m is defined as in Lemma 7. This, in combination with item 2 of Lemma 8 (for almost all θ , $\sup_{\lambda \in \Lambda} |SURE(\lambda, \theta') - SURE(\lambda, \theta)| \rightarrow 0$ as $\theta' \rightarrow \theta$), implies that $\tilde{\lambda}(\theta', \Delta)$ is such that $g^{\tilde{\lambda}(\theta', \Delta)}(\theta') = -\theta'$ for all (θ', Δ) in a neighborhood of $(\theta, 0)$, and the claim follows. $\square$

F.2 Proof of convergence in distribution

We can now prove Lemma 5, drawing on our preceding Lemmas.

Proof of Lemma 5:

- By Lemma 2,

$$\hat{\theta}_n \rightarrow^d \hat{\theta} \sim N(\theta_0, \Sigma).$$

- Let $\Delta_n(\lambda) = CV_n(\lambda) - c_n - SURE(\lambda, \hat{\theta}_n, \Sigma)$. By Lemma 4,

$$\sup_{\lambda \in \Lambda} |\Delta_n(\lambda)| \rightarrow^p 0.$$

- By joint convergence (van der Vaart 2000, item (v) of Theorem 2.6), $W_n = (\hat{\theta}_n, \Delta_n) \rightarrow^d W = (\hat{\theta}, 0)$.
- By Lemma 9, the mapping from W_n to $\hat{\theta}_n + g^{\tilde{\lambda}(\hat{\theta}_n, \Delta_n)}(\hat{\theta}_n)$ is almost surely continuous on the support of W .
- By definition, $\tilde{\lambda}(\hat{\theta}_n, \Delta_n) = \lambda_n^* = \operatorname{argmin}_{\lambda} CV_n(\lambda)$, and $\tilde{\lambda}(\hat{\theta}, 0) = \lambda^* = \operatorname{argmin}_{\lambda} SURE(\lambda, \hat{\theta}, \Sigma)$.
- The almost surely continuous mapping theorem (van der Vaart 2000, Theorem 2.3) then implies

$$\hat{\theta}_n + g^{\lambda_n^*}(\hat{\theta}_n) \rightarrow_d \hat{\theta} + g^{\lambda^*}(\hat{\theta}) = \hat{\theta}^*.$$

- By Lemma 2 $\hat{\theta}_n^* = \hat{\theta}_n + g^{\lambda_n^*}(\hat{\theta}_n) + o_p(1)$, and the claim follows.

□

F.3 Convergence of loss and risk

Proof of Theorem 1:

By Lemma 3,

$$\bar{L}_n(\theta, \theta_0) \rightarrow \frac{1}{2} \|\theta - \theta_0\|^2$$

uniformly in any bounded neighborhood of θ_0 . By Lemma 5,

$$\hat{\theta}_n^* \rightarrow_d \hat{\theta}^*.$$

Combining these two results gives

$$\bar{L}_n(\hat{\theta}_n^*, \theta_0) \rightarrow_d \frac{1}{2} \|\hat{\theta}^* - \theta_0\|^2.$$

The distributional convergence claim of Theorem 1 follows. The claim of Corollary 1 is then immediate. $\square$